

Subjective versus Objective Incentives and Employee Productivity^{*}

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Abstract

Incentivizing employees is challenging: objective incentives may distort effort toward measurable outcomes, while subjective (manager-discretionary) incentives depend on accurate evaluation. We study both relative to no incentives for teachers in an RCT across 200+ Pakistani schools. Subjective and objective incentives produce similar test score gains, but objective incentives reduce non-test student outcomes and change teaching practices. We rationalize these effects through the lens of a moral hazard model with multi-tasking and show teachers believe objective incentives are a noisier function of their effort than subjective incentives, and objective incentives distort effort toward narrow aspects of student learning.

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1 Introduction

How should firms incentivize employees when effort is non-verifiable or non-contractable? Contract theory provides an answer. The second best is to incentivize outcomes of the employee’s production function. However, this introduces two new problems – distortion (over-incentivizing measurable outcomes while ignoring others) and noise (outcomes are a noisy function of employee effort). How do most firms actually incentivize workers? They use manager-discretionary (subjective) incentives rather than outcome-based (objective) ones. In the US, 85% of full-time employees have at least one aspect of their compensation determined by their manager, and 90% of teacher performance evaluations have a subjective component (Engelland and Riphahn, 2011; National Center for Education Statistics, 2011; Frederiksen et al., 2017). Despite the prevalence of subjective incentives, causal evidence on their effects remains limited (Rockoff and Speroni, 2010). These issues are particularly salient in sectors which exhibit a complicated production function, with many difficult to quantify outcomes, such as teaching (Jackson, 2018).

In this paper, we ask two questions: What is the effect of subjective versus objective incentives versus no incentives on teacher productivity? To what extent can contract noise and distortion explain the observed response to subjective versus objective incentives? We answer these questions by conducting an 18-month randomized controlled trial with 234 private schools in Pakistan. We randomize schools to provide teachers with one of three contracts: (i). control: all teachers receive a raise of 5% irrespective of performance, (ii). subjective treatment: teachers receive a raise from 0-10% based on their manager’s rating of their performance, or (iii). objective treatment – teachers receive a raise from 0-10% based on their students’ mid-year and end-of-year percentile value-added (Barlevy and Neal, 2012). Both treatments are within-school tournaments and have the same distribution of raise thresholds. These similarities allow us to isolate the effort response from just changing the performance metric (manager rating versus test score value-added) while holding other features of the incentive structure constant.

We use detailed administrative, survey, test, and classroom observation data to understand each contract’s effect on teacher effort and student outcomes. Student outcomes are measured along two dimensions: test scores and socio-emotional development. Test score data comes from an endline test conducted by the research team, one month after the end of the contract. Students are tested in core subjects (English, Urdu, math, science, and economics) in grades 4-13. A variety of question types and sources allow us to test whether effects are driven by memorization-type questions. Socio-emotional development is measured along six dimensions: love of learning, ethical decision-making, inquisitiveness, global citizenship, sustained attention and resilience. The first four dimensions are measured using student self-report on items drawn from standard psychological indices of children’s socio-emotional development. The latter two dimensions are measured using within-item variation in performance on the academic subject tests.

In our first main result, we show that both subjective and objective contracts are equally effective

at increasing test scores. Both contracts increase test scores by 0.09sd, which is very similar to average effects from meta-analyses of performance pay for teachers (Pham et al., 2020). These results are consistent across subject and grade and are not driven by rote-memorization type questions. However, in contrast to the test score results, objective and subjective incentives have different effects on socio-emotional skills. Objective incentives have a slight negative effect on student socio-emotional development, significantly decreasing love of learning and increasing students’ desire to change schools. Subjective incentives result in a small positive effect on overall socio-emotional skills. We also find that students in the subjective incentive schools show higher levels of resilience and sustained attention than those in objective incentives and control schools. These skills are measured by comparing performance on test items directly after a difficult item (perseverance) or later in the exam (sustained attention). These combined effects suggest that teachers under objective contracts focused on improving student test performance, at the cost of more well-rounded development for students. Whereas, teachers under the subjective contract were able to prioritize both areas.

To understand teachers’ behavioral responses to these incentive contracts, we compile rich data on teacher behavior inside and outside the classroom. We record 6,800 hours of classroom footage and review it using a standard classroom observation rubric (Pianta et al., 2012). The rubric captures teacher behavior along a dozen dimensions, from the use of punitive discipline to the proportion of student versus teacher talk time. The rubric also measures the amount of time spent on test-taking or test-preparation activities. To measure effort outside instructional time, we acquire administrative data on daily clock in and out time and combine with teacher self-reported time use. Together these data sources allow us to understand teacher behavior change under subjective versus objective incentives.

In our second main result, we find both subjective and objective incentives lead to changes in classroom practices. As one might expect, subjective incentives spur actions that managers value, and objective incentives spur actions that most quickly and easily translate into test score gains. Subjective incentives lead to increased teacher attendance (relative to the control group) and improved pedagogy in some areas (relative to the objective treatment), like focusing on different student needs. Objective incentive schools see a five-fold increase in class time on test preparation activities, more lecturing compared to student-centered teaching and a more negative classroom climate (with increases in teacher yelling and use of punitive discipline).

The results on student outcomes and teacher behavior suggest that subjective performance incentives changed teacher effort or priorities without producing distortionary effects. How are managers able to accomplish this? To understand what managers valued and how they evaluated teachers we use survey data of teacher’s beliefs, manager vignette responses and analyze the text of the criteria managers used for evaluations. We find, on average, managers tend to value the actions which contribute to both test-score and non-test score outcomes for students. We also do not find evidence that teachers believe subjective incentives are more susceptible to favoritism or bias. However, there is heterogeneity in how effectively managers implement the contract. In particular,

for managers whom teachers believe are poorly informed about employee performance, we cannot reject that subjective incentives have no effect on student outcomes relative to the control.

The implication of these results is not that subjective incentives are universally more effective than objective incentives, but in this context with the particular features of these subjective and objective incentives, the former appears to produce the same test score results as the latter without any of the detrimental consequences shown by the objective arm. To help us better understand why this version of subjective incentives appeared to work better, we construct a model starting from the classic moral hazard model with multi-tasking as presented in Baker (2002). We then restrict to a specific class of incentive contract types and show two key predictions of the model which we test experimentally: worker productivity is decreasing in incentive contract *noise* (lack of correlation between employee action and incentive pay) and *distortion* (1 - correlation between the piece rate for different actions and marginal return to those actions on firm outcomes).

Rather than claiming that one incentive type universally dominates, we use the experimental variation to quantify how noise and distortion — features that vary across the two treatments — shape the relative effectiveness of each approach. To test whether these two mechanisms, noise and distortion, can explain our reduced form results, we take the following approach: First, we show that employees believe the subjective and objective contracts are similar along all dimensions except the extent of noise and distortion. Second, we exploit partially exogenous heterogeneity *within* a given treatment to isolate the causal effect of noise and distortion on student outcomes. Finally, we bring those two estimates together to show that given the difference in perceived noise and distortion across the contracts and the causal effect of noise and distortion on student outcomes, we can explain a large portion of the reduced form effects with these two channels. We explain each step in detail below.

The first step is testing whether teachers believe there are differences in the extent of noise and distortion across the two treatments. We do this by asking teachers at endline to what extent working harder would increase their incentive pay and what types of actions (lesson planning, improving pedagogy, helping other teachers, etc) are rewarded under each system. We find that teachers believe subjective performance incentives are *less* noisy than objective incentives and view subjective incentives as more effective at motivating effort. They view the objective score as less within their control as many other factors beyond their effort affect student test scores. Teachers in the objective treatment are more likely to prioritize the type of actions which lead to test score gains, at the cost of other areas of student development. Teachers under subjective contract prioritize actions that lead to test score and non-test score gains and also prioritize administrative tasks, which are likely to be valued by managers. We also show there are no other perceived differences across the two treatment arms, including similarity in implementation timelines, understanding of the contract treatments, and beliefs about the fairness of each treatment arm.

To determine the causal effect of noise on student outcomes, we need variation in how closely teachers' effort maps into incentive pay, holding all else equal. To test this, we exploit the fact

that some managers are perceived to have very accurate subjective evaluations, whereas others are not well-informed about their subordinates' effort. We explicitly measure this by asking teachers to rate how accurate their manager is in rating *other* teachers. All managers, irrespective of school treatment status, have to provide an annual subjective evaluation score for each teacher, so we can construct this measure for all managers. We then use this accuracy measure X subjective treatment as our instrument for contract noisiness. The idea being that manager inaccuracy will be correlated with teachers' perceptions of how noisy the contract is only in subjective schools and should not affect their perception of how noisy the contract is in objective or control schools.

Using the interaction of manager inaccuracy x subjective treatment to instrument for noise, we find that a 1 sd increase in the perceived noisiness of the contract decreases student test scores by 0.2 sd. This suggests that employees' effort response is very sensitive to the noisiness of the contract. The primary exclusion restriction concern is that inaccurate managers could implement the subjective incentive scheme differently in other ways (for example, maybe they focus on different kinds of pedagogy). We show our results are robust to a variety of controls for subjective evaluation criteria selected and contract knowledge accuracy.

Next, to isolate the causal effect of contract distortion on student outcomes, the ideal experiment would be to randomly vary what teachers are told to focus on (e.g. lesson plan preparation vs. organizing after school activities, etc) and measure the effect on student learning. We cannot implement that idealized design, but we can exploit the fact that different managers prioritized different actions in evaluating teachers. We use data on the weight allocated to each evaluation criteria and use the text of the criteria to categorize them into five types of teacher actions: administrative tasks, professional development tasks, improvements in teacher pedagogy, test-score based goals and other. All managers have to delineate and score teachers on criteria, but in subjective treatment schools financial incentives are then tied to these criteria.

Of course, schools in which managers focus on administrative goals versus those in which managers focus on pedagogy goals are likely different in many ways. Therefore, we interact the weight put on each action category with subjective treatment status. In effect, we are comparing schools in which principals have delineated the same evaluation criteria but are either randomized to subjective (financial stakes of these evaluation criteria) versus objective and control (no financial stakes of these evaluation criteria).

We find that a larger focus on test scores and professional development increases students' endline test scores. However, more focus on test scores results in negative effects on student socio-emotional development. Again, the key exclusion restriction violation would be that managers who prioritize certain evaluation criteria also implement the subjective incentives differently in other ways. Our findings are once again robust to adding in controls for perceived contract noise and contract knowledge accuracy.

Combined, these results help us understand why it is possible for both subjective and objective incentives to have the same effect on test scores even though only one of them incentivized test scores

directly. Subjective incentives are less noisy, producing a larger overall effort response, and prioritize both test score and non-test score student outcomes, allowing teachers to prioritize multiple areas of student development. We find that the noise and distortion channels are able to explain a substantial fraction of the reduced form effects we see across test scores and socio-emotional outcomes.

Our paper makes three key contributions. First, it is the first study, to our knowledge, to isolate the causal effect of purely subjective versus objective incentives and the effect of purely subjective versus flat pay for employees in any sector (Lazear and Oyer, 2012; Oyer and Schaefer, 2011). Existing experimental studies have tested bundled incentives (e.g. combined subjective and objective incentives) versus no incentives on employee behavior (Bandiera et al., 2020; Khan et al., 2019; Fryer, 2013; Leaver et al., 2021) or varied the characteristics within a subjective evaluation system (de Janvry et al., 2023). Previous work has also compared the effect of heterogeneity across plants to measure the effect of more or less steep subjective incentives on employee overtime (Engelland and Riphahn, 2011). There is also evidence that managers, especially in educational settings, may have imperfect information about worker effort or may be biased toward certain groups (Eren, 2023; Gibbs et al., 2004; Jacob and Lefgren, 2008).

Second, we add to a robust literature on the effect of performance pay for teachers by providing two new findings (Biasi, 2021; Fryer, 2013; Goodman and Turner, 2013; Lavy, 2007; Muralidharan and Sundararaman, 2011). We show the first evidence of objective performance pay having detrimental effects on non-academic student outcomes, consistent with multi-tasking models. Next, we show direct evidence that objective incentives result in teachers distorting their effort toward teaching pedagogy that impacts test performance at the cost of other areas of student development. This includes the use of class time doing test prep and the use of punitive discipline. Both of these results have long been suspected, but we provide the first documentation of such effects (Baker, 2002; Deserranno et al., 2025; Leigh, 2013).

Third, we provide, what we believe is, the first evidence on measuring the extent of noise and distortion within an employee’s contract and isolating the effects of those mechanisms on firm outcomes. There is a rich theoretical literature on the importance of these mechanisms (Baker, 2002; Yang, 2008) and some empirical work has also investigated the role of noise on employee response (Prendergast, 1999; Prendergast and Topel, 1993; Prendergast, 2007) in non-experimental settings.

The remaining sections are organized as follows. Section 2 gives an overview of the standard moral hazard model with multi-tasking and highlights the two key mechanisms which underpin the reduced form effects we find. Section 3 details the treatment and control conditions, the data collected, and standard implementation checks. Section 4 provides the main results of subjective and objective performance incentives on teacher effort and student outcomes. Section 5 unpacks the mechanisms underlying the main effects in light of the moral hazard model, and section 6 concludes.

2 Conceptual Framework

To understand the potential effects of subjective versus objective incentives, we begin with a simple model of moral hazard with multitasking as presented in Baker (2002), which provides a very general framework for how incentives may affect employee behavior. We will then restrict to a subset of performance contract types to further tighten our predictions about the effect of each contract, beyond Baker (2002). At the core of this model is the tension between paying employees for the outcomes the firm cares about while not penalizing employees for things outside of the employee's control. This is also the key tension at the heart of the costs and benefits of using subjective versus objective incentives, which we outline in section 2.3. Appendix C extends the framework to match more closely with the features of the contract in the experimental design and shows the key comparative statics from the simple model hold in the more complex version.

2.1 Performance Pay Contracts

The firm, a school, produces a single outcome – student learning, $V(\mathbf{e}, \epsilon)$ – through a simple linear production function:

$$V(\mathbf{e}, \epsilon) = \mathbf{f} \cdot \mathbf{e} + \epsilon = f_1 e_1 + f_2 e_2 + \dots + \epsilon \quad (1)$$

Student learning is a function of an n -dimensional vector of teacher effort choices, $\mathbf{e}$, and the n -dimensional vector of marginal products of those actions, $\mathbf{f}$. The n dimensions of effort are concrete daily choices teachers make – days present at school, minutes per class spent on student-centered pedagogy, minutes spent on test preparation, time spent planning lessons, and so on. Student learning is also a function of many things outside the teacher's action set (environment, parental support, peers, etc.), which are captured by the noise term, ϵ , which is mean zero with variance σ_ϵ^2 .

Employers cannot perfectly observe effort, but they can generate a performance measure, $P(\mathbf{e}, \phi)$, which is some combination of observable outputs (test scores, student attendance, etc.) and/or observable teacher actions (attendance, lesson plans, etc.):

$$P(\mathbf{e}, \phi) = \mathbf{g} \cdot \mathbf{e} + \phi = g_1 e_1 + g_2 e_2 + \dots + \phi \quad (2)$$

Here $\mathbf{g}$ is the vector of marginal returns of each effort dimension to the performance measure, and ϕ captures everything that shifts P independently of $\mathbf{e}$. ϕ is mean zero with variance σ_ϕ^2 . Two components are captured by ϕ : (i) factors outside the teacher's control that affect the measure (e.g., students' home environment under test-score value added; idiosyncratic variation in the manager's mood or attention under subjective evaluation) and (ii) mis-measurement of teacher actions (e.g., imperfect attendance records, or the gap between the teacher's own belief about her effort and the principal's noisy perception of it).

The actual experimental design features a within-school tournament with five rank-based categories unlike the simple linear piece rate. In appendix B, we provide empirical evidence for the linear piece rate simplification and show teachers beliefs and actions are consistent with this simplified model. Then in appendix C, we augment this simple model to feature a multi-rank category tournament system and show the two main comparative statics also hold under such a contract design.

2.2 Teacher's problem and equilibrium effort

Teachers have quadratic costs of effort around a baseline action vector $\vec{\eta}_i$ they would choose absent incentives:¹

$$u(\mathbf{e}, \phi) = \mathbf{g} \cdot \mathbf{e} + \phi - \sum_{i=1}^n \frac{(e_i - \eta_i)^2}{2} \quad (3)$$

Teachers are heterogeneous in $\vec{\eta}_i$ (their no-incentive baseline effort) but not in their cost of effort. $\vec{\eta}_i$ is the action vector teacher i would choose in the control arm of the experiment, and can reflect cultural, professional, or institutional norms that vary across teachers. We do not assume $\vec{\eta}_i$ is fixed across treatments at the individual level. We only assume (and test in figure A1) that its cross-sectional variance within a school is similar across treatments.

Taking the first order condition of eq. (3),

$$e_{ik}^* - \eta_{ik} = g_k, \quad (4)$$

so equilibrium effort is the baseline plus the effective piece rate. Define $\Delta e_{ik}^* \equiv e_{ik}^* - \eta_{ik}$ as the effort response relative to no incentive pay. Expected student learning relative to the no-incentive baseline is then:

$$\Delta V_i^* \equiv \mathbb{E}[V(\mathbf{e}_i^*, \epsilon)] - \mathbb{E}[V(\vec{\eta}_i, \epsilon)] = \mathbf{f} \cdot \mathbf{g} = \|\mathbf{f}\| \|\mathbf{g}\| \cos \theta, \quad (5)$$

where θ is the angle between vectors $\mathbf{f}$ and $\mathbf{g}$. Average student learning then is a function of the length of the marginal production on student learning vector, $\|\mathbf{f}\|$, the length of the piece-rate vector, $\|\mathbf{g}\|$, and the alignment between these two vectors, $\cos(\theta)$. In other words, student learning is increasing in the steepness of the incentives and how aligned those piece rates are with the student learning production function.

Departing from Baker (2002), we can further re-arrange the expression to show the effect that noise in the performance measure has on average student learning. Taking the variance of eq. 2

¹Baker (2002) assumes risk-averse agents with utility $u(\mathbf{e}, \phi) = E[P] - r \text{var}[P] - \sum_i e_i^2/2$. Because we are not studying teacher retention, we omit the risk-aversion term, which only enters the participation constraint and does not change the effort first order condition for a teacher who has accepted the contract.

and assuming the covariance matrix of $\vec{e}$ is proportional to the identity matrix, we have $\text{var}(P) = \|\mathbf{g}\|^2 \text{var}(\|\mathbf{e}\|) + \sigma_\phi^2$. Re-arranging, we can substitute this in for $\|\mathbf{g}\|$ into eq. 5. Average student learning then is:

$$\Delta V_i^* = \|\mathbf{f}\| \frac{\sqrt{\text{var}(P) - \sigma_\phi^2}}{\sqrt{\text{var}(\|\mathbf{e}\|)}} \cos\theta \quad (6)$$

Here $\text{var}(\|\mathbf{e}\|)$ is the cross-teacher variance in the norm of the effort vector within a school, which captures how much teachers within a school differ in total effort. It does not require that total effort be fixed in the absence of incentives. Our assumption that this quantity is similar across treatments follows from assuming homogeneous effort-cost slopes (teachers differ in $\vec{\eta}_i$ but respond to a given g_k in the same way), and we test it directly in the data by comparing the within-school variance of 25 measured effort dimensions across treatment arms.²

Our experimental design also fixes $\text{var}(P)$ across the two treatment arms: the prize schedule is identical, and by construction the rank distribution (and therefore the cross-teacher variance of realized compensation) is the same in both arms. This is a feature shared by essentially any tournament or relative-performance system with a fixed prize structure.

2.3 Theoretical Predictions

With $\|\mathbf{f}\|$, $\text{var}(P)$, and $\text{var}(\|\mathbf{e}\|)$ equal across the two treatments, the reduced-form difference in ΔV^* between the subjective and objective arms is driven by just two channels – the **noise** σ_ϕ^2 and the **distortion** $1 - \cos\theta$:

- $\frac{\partial \Delta V^*}{\partial (1 - \cos\theta)} < 0$: Average student learning is decreasing in incentive **distortion**, $1 - \cos(\theta)$
- $\frac{\partial \Delta V^*}{\partial \sigma_\phi^2} < 0$: Average student learning is decreasing in incentive **noise**, σ_ϕ^2

Distortion Distortion captures the correlation between the piece rates for different actions and the marginal return to student learning of those actions. In essence, do we pay teachers more for the actions which are more related to developing student learning? The more distorted a contract is, the more employees focus on actions that are less helpful toward student learning.

Noise Noise captures how much of the performance incentive is unrelated to employee’s actions. This could operationalize as other factors outside the employee’s control affecting the performance measure (school resources, shocks, etc.) or mis-measurement of employee actions, if the contract attempts to measure teacher actions. It is important to flag that traditionally the way noise enters

²We calculate the within-school variance in effort along 25 measured dimensions of teacher effort. Figure A1 plots the distribution of p-values for F -tests of equality across each treatment arm pair. The distribution is approximately uniform on $[0, 1]$, providing supporting evidence that $\text{var}(\|\mathbf{e}\|)$ is constant across treatment.

the optimal contract design is through reducing risk-averse employee’s utility. This requires firms to raise the fixed part of an employee’s salary to meet employee’s participation constraint. Here we are not focused on that additional consequence of noise as we are not focused on employee entry or exit in this paper.³

The effect of noise we focus on here is equivalent to a decrease in the incentive scheme’s average piece rate. Since $\sigma_\phi^2 = \text{var}(P) - \|\mathbf{g}\|^2 \text{var}(\|\mathbf{e}\|)$, and $\text{var}(P)$ and $\text{var}(\|\mathbf{e}\|)$ are constant given the tournament nature of each incentive scheme, increasing σ_ϕ^2 directly decreases $\|\mathbf{g}\|$. Therefore, increasing noise then reduces the extent of the effort response, $\|\mathbf{e}^*\|$. This effect of noise exists in any incentive scheme with a fixed variance, which includes all tournament or threshold-type incentives.

Subjective vs. Objective Incentives We would expect that both subjective and objective incentives will exhibit some level of noise and distortion. The sources of noise and distortion for each incentive can be summarized as the following:

Distortion	Subjective	- Gap between manager’s utility function and student learning - Incorrect beliefs about teacher’s production function
	Objective	- Gap between objective incentive metric and student learning - Incorrect beliefs about own production function
Noise	Subjective	- Mismeasurement of effort - Components of manager’s utility function outside teacher’s control
	Objective	- Mismeasurement of objective incentive metric (i.e. test scores) - Components of objective incentive metric outside teacher’s control

The theoretical framework allows us to understand how a given incentive metric, P , may affect teachers’ responses and, as a result, student learning. Ex-ante, it is not clear whether subjective or objective incentives would be more or less distorted in the teaching context. On the one hand, subjective incentives may solve the multi-tasking problem by prioritizing more than just test scores. One of the key critiques of objective incentives is that teachers may focus on actions which enhance test scores (such as test prep skills, memorization, etc.), but have small or zero effects on true human capital (Muralidharan and Sundararaman, 2011). Subjective performance incentives would ideally penalize these types of behaviors and reward more well-rounded teaching. On the other hand, managers may prioritize the wrong actions – because they do not know what the true teacher production function is, because they value aspects of student learning differently than the firm, or, most nefariously, they weight actions which contribute solely to their personal utility (rewarding friends, offloading their own tasks, etc).

It is also uncertain whether subjective or objective would be less noisy. Test scores are notoriously noisy measures of teacher effort (Chetty et al., 2014). One of the most common

³A companion paper (Brown and Andrabi, 2025) studies employee selection in response to these contracts.

complaints teachers have against test score-based incentives is that they are mostly unrelated to teacher actions (Podgursky and Springer, 2007). Subjective performance pay could be less noisy than objective performance pay because managers could focus on rewarding actions rather than outcomes. However, this requires managers to observe effort accurately. Subjectivity could introduce additional noise if managers introduce bias or favoritism into their evaluations.

3 Experimental Design

3.1 Connecting Framework and Experiment

We first seek to understand the effect of different performance pay schemes on student learning, V , and teacher effort, $\vec{e}$, in section 4. This section will detail the different versions of performance pay, P , and our measurement of V and $\vec{e}$. Then, in section 5, we will test whether we can in fact explain differences in the effect of P through differences in the level of noise, σ_ϕ^2 , and distortion, $1 - \cos(\theta)$.

3.2 Performance Incentive Treatments

We partnered with a private school network of 234 schools in Pakistan to implement the research design. Schools are randomized to receive one of three contracts which determine the size of teachers' raises at the end of the calendar year.⁴ The three contracts were:

- **Control: Flat Raise** - Teachers receive a flat raise of 5% of their base salary
- **Treatment: Performance Raise** - Teachers receive a raise from 0-10% based on their within-school performance ranking using the percentile thresholds below:

Performance Group	Within-School Percentile	Raise amount
Significantly above-average	91-100th	10%
Above-average	61-90th	7%
Average	16-60th	5%
Below average	3-15th	2%
Significantly below average	0-2nd	0%

There are two treatment arms, which vary what performance metric is used to evaluate teachers. Teachers in a given treatment arm are ranked within their school on one of the following performance measures:

⁴Triplet-wise randomization by baseline test performance was used, which generally performs better than stratification for smaller samples (Bruhn and McKenzie, 2009).

- ***Subjective Treatment Arm:*** Teachers are evaluated by their manager at the end of the calendar year. Managers had complete discretion over how they evaluated teachers and what aspects of performance they would prioritize. To ensure teachers knew what was expected of them, managers delineated between 4-10 evaluation criteria at the beginning of the year. At the end of each evaluation period, managers scored teachers on each criterion, producing a composite score that directly determined the teacher’s within-school ranking and corresponding raise category. These varied substantially by teacher and manager and included criteria such as improving their behavioral management of students, assisting with administrative tasks, helping plan an afterschool event, improving students’ spoken English proficiency, etc. An example set of criteria are provided in figure A2 and a summary of how common different criteria were across our sample is shown in figure A3.
- ***Objective Treatment Arm:*** Teachers are evaluated based on their average percentile value-added (Barlevy and Neal, 2012) for the spring and fall semester. Percentile value-added is constructed by calculating students’ pre-semester percentile for a given subject-grade within the entire school system and then ranking their of post-semester score relative to all other students who were in the same pre-semester percentile.^{5,6} We then average across all students the teacher taught during the two terms.

The contract applied to all core teachers (those teaching Math, Science, English, and Urdu) in grades 4-13. Elective teachers and those teaching younger grades received the status quo contract. All three contracts have similar budgetary implications for the school. We over-sampled the number of subjective treatment arm schools due to partner requests, so the ratio of schools is 4:1:1 for subjective treatment, objective treatment, and control, respectively.

Both the subjective and objective treatment arms have several features in common, allowing us to isolate the effect of differing the performance metric while keeping all other features constant. Both treatments are within-school tournaments, so this holds the level of competition fixed between the two treatments. In addition, the variance in the distribution of the incentive pay is equivalent across the two treatments. As we showed in section 2 since the variance in effort is constant, holding the variance in P constant allows us to interpret differences in noise levels between the two systems as equivalent to differences in incentive steepness. The performance evaluation timeline also played out the same for all groups. Before the start of the year, managers set performance goals for their teachers irrespective of treatment. Teachers were evaluated based on their performance in January through December, with testing conducted in June and January to capture student learning in each

⁵Existing school system wide assessments are used for the pre-semester ranking for spring and fall quarters and post-semester spring. Assessments conducted by the research team are used for the post-semester fall scores.

⁶Percentile value-added has several advantageous theoretical properties (Barlevy and Neal, 2012) and is also more straightforward to explain to teachers than more complicated calculations of value-added.

term of the year.⁷

To ensure teachers and managers had full understanding of how each contract would work, we conducted an intensive information campaign with schools. First, the research team had an in-person meeting with each manager, explaining the contract assigned to their school, and, in the case of the subjective treatment, explaining what would be expected of them and when. Second, the school system’s HR department conducted in-person presentations once a term at each school to explain the contract. Third, teachers received frequent email contact from school system staff reminding them about the contract and half-way through the year contract teachers were provided midterm information about their rank based on the first 6 months.⁸ Control teachers were also provided information about their performance in one of the two metrics, in order to hold the provision of performance feedback constant across all teachers.

Other aspects of the performance management process remain fixed across all three experimental arms. Managers maintain authority to fire low performing employees (a rare occurrence) and higher level school system staff decide on promotions with guidance from the teacher’s manager.

3.3 Timeline and Data

Our study was conducted from October 2017 through June 2019. It covered one performance review cycle conducted from January-December 2018 during which the contracts were in place. Figure 1 presents the main treatment implementation (detailed in section 3.2) and data collection activities (detailed below).

Our data allows us to understand how teachers changed their effort under each incentive scheme, why the incentives affected effort in the way they did, and the resulting effect this had on student outcomes. We draw on data from (i). the school system’s administrative records, (ii). baseline and endline surveys conducted with teachers and managers (iii). endline student testing and survey and (iv). detailed classroom observation data.

Administrative Data: The administrative data details position, salary, performance review score, attendance, and demographics for all employees. We also have biometric clock in/out data for all schools. The data was provided by the school system for the period of July 2016 to June 2019. It includes classes and subjects taught for all teachers, and end of term standardized exam scores for

⁷The school system’s central office designed and administered the June test to all students in a given grade. However, tests are graded locally by the school, often by the students’ teacher. Due to concerns of grade manipulation, grading was audited by the research team. 10% of all teacher’s exams were regraded. If the teachers’ grade and the auditor’s grade were off by more than 5%, another 10% of their tests were audited. If the average was still off by more than 5%, all of the teacher’s exams were regraded. Overall, grade manipulation was small and was generally driven by cases where teachers bumped up students’ grades from just failing to just passing. There was no heterogeneity in grading accuracy by treatment. These tests are *not* used as an outcome measure in this paper. The January test was conducted exclusively by the research team (described in section 3.3 below) and therefore this is the test we use to measure learning effects.

⁸An example midterm information note is provided in Appendix Figure A4.

all students (linked to teachers). From September through December 2018, we also have data on classroom observations conducted by managers. Managers use a similar rubric to the one used by the research team to conduct classroom observations (detailed below).

Baseline Survey: The baseline survey measured teachers’ preferences over different contracts and beliefs about their performance under each contract. 40% of schools were randomly selected to participate in an in-person baseline survey conducted in October 2017. 2,500 teachers and 119 managers were surveyed. These outcomes are primarily used for a companion paper on teacher selection in response to performance pay (Brown and Andrabi, 2025).

Endline Survey: The teacher endline survey measured their understanding of the assigned contract, time use, and beliefs about their manager’s level of bias in conducting performance evaluations. The manager endline survey measured managers’ beliefs about teacher quality and measured management quality using the World Management Survey school questionnaire.⁹ The endline survey was conducted online with teachers and managers in spring and summer 2019. 6,080 teachers and 189 managers were surveyed.

Endline Student Testing and Survey: An endline test was conducted with students to measure performance in core subjects and socio-emotional skills after one year of the intervention. The research team conducted the endline test and student survey in January 2019. The test was conducted in Reading (English and Urdu), Math, Science, and Economics. The items were written in partnership with the school system’s curriculum and testing department to ensure appropriateness of question items. Grading was conducted by the research team. Items from international standardized tests (PISA, TIMSS, PERL, and LEAPS) and a locally used standardized test (LEAPS) were also included to benchmark student performance.¹⁰

Students also completed a survey to measure five areas of socio-emotional development: love of learning, ethical decision-making, global citizenship, inquisitiveness and attachment to their school. The choice of these five areas came from the school system’s priorities. These values are posted on the walls in schools, and teachers receive professional development on building these values in students. Some managers also specifically refer to development of these student skills in teachers’ evaluation criteria.

We measure improvement in these areas using student responses to items drawn from several validated, age-appropriate, psychometric surveys which aim to measure these skill areas.¹¹ These

⁹Due to budget constraints, we were unable to have the World Management Survey surveyors conduct the survey. Instead, we asked managers to directly rate themselves on the rubric that surveyors use. This approach could result in inflated management scores. As a result, we corroborate managers self reports with teacher survey and administrative data on observations completed.

¹⁰The endline student test data was used both for evaluating the effect of the treatments and used to compute objective treatment teachers’ raises.

¹¹Items are drawn from the Achievement Emotions Questionnaire, Learning and Study Strategies Inventory,

items have been shown to correlate with student motivation, high school graduation, college attendance and civic participation (Amoateng et al., 2014; Kashdan et al., 2018; Pekrun et al., 2002; Tang and Salmela-Aro, 2021; Van Wyk and Mason, 2021). Appendix table A1 lists the survey items used for each area along with their source.

Finally, we measure two additional aspects of student executive functioning, resilience and sustained attention, using performance on certain aspects of the academic tests. Question order is randomized on academic tests, so we measure student resilience by comparing student performance on question items that appear just after very difficult items. We measure sustained attention by comparing performance on earlier versus later question items on the exam.¹²

Classroom Observation Data: To measure teacher behavior in the classroom, we recorded 6,800 hours of classroom footage and reviewed it using the Classroom Assessment Scoring System, CLASS (Pianta et al., 2012), which measures teacher pedagogy across a dozen dimensions.^{13,14,15} We also recorded whether teachers conducted any sort of test preparation activity and the language fluency of teachers and students.

Performance Evaluation Data: We also draw on data from the school system’s existing performance evaluation system. Prior to and during the intervention, the school system maintained a performance management system in which managers would delineate 4-10 evaluation criteria and the weight attached to each criteria. These criteria are open fields for managers decide on for each of their employees. We use the open response text and point weightage to understand what aspects of teacher behavior managers placed weight on. Managers select these criteria in December 2017 before treatment status was announced.

Eisenberg’s Child-Report Sympathy Scale, Bryant’s Index of Empathy Measurement, McCabe/ICAI Academic Integrity Survey, Afrobarometer, World Values Survey, and Epistemic Curiosity Questionnaire.

¹²These two latter dimensions of socio-emotional development were suggested after data collection based on seminar feedback and therefore were not preregistered.

¹³Classroom observations were conducted in a randomly selected subset of 60% of the sample schools due to budget constraints.

¹⁴There are tradeoffs between conducting in-person observations versus recording the classroom and reviewing the footage. Videotaping was chosen based on pilot data which showed that video-taping was less intrusive than human observation (and hence preferred by teachers). Videotaping was also significantly less expensive and allowed for ongoing measurement of inter-rater reliability (IRR).

¹⁵We did not hire the Teachstone staff to conduct official CLASS observations as it was cost-prohibitive and we required video reviewers to have Urdu fluency. Instead we used the CLASS training manual and videos to conduct an intensive training with a set of local post-graduate enumerators. The training was conducted over three weeks by Christina Brown and a member of the CERP staff. Before enumerators could begin reviewing data, they were required to achieve an IRR of 0.7 with the practice data. 10% of videos were also double reviewed to ensure a high level of ICC throughout the review process. We have a high degree of confidence in the internal reliability of the classroom observation data, but because this was not conducted by the Teachstone staff, we caution against comparing these CLASS scores to CLASS data from other studies.

3.4 Sample and Characteristics of the Employee-Manager Relationship

Teachers The study was conducted with a large private school system in Pakistan. Table 1 presents summary statistics of our sample compared to a representative sample of teachers in US (National Center for Education Statistics, 2011). The student body is from an upper middle-class and upper-class background. Teachers are generally younger and less experienced than their counterparts in the US, though they have similar levels of education. Our sample is mostly female (80%), young (35 years on average), and inexperienced (5 years on average, but a quarter of teachers are in their first year teaching). All teachers have a BA and 68% have some post-BA credential or degree.

Managers In order to understand the effects of subjective performance pay, we need to understand who the managers are and what role they play in overseeing teachers. Managers here are either a principal in small schools or a vice principal in larger schools. They are tasked with overseeing the overall operations of the school and managing employees, including teachers and other support staff. Table 2 presents information about managerial duties compared to a US sample of principals. Like in the US, these managers are generally older (45 years old), less likely to be female (61%), and more experienced (9.6 years) than teachers. Most were previously teachers and transitioned into an administrative role.

The distribution of time use is fairly similar to the principals in the US. Managers spend about a 1/3 of their working hours overseeing their staff – observing classes, providing feedback, meeting with teachers and reviewing lesson plans. The rest of their time is spent on other tasks related to the schools functioning. However, managers in our sample spend much more time directly observing teachers. They do about twice the number of classroom observations each year (4.7 versus 2.5 in the US). They also rate themselves higher in most areas of the management survey questions (4.3 versus 2.8 out of 5), including formal evaluation, monitoring and feedback systems for teachers. This is an important difference as these management practices could positively affect the success of the subjective treatment arm, and may inform the extent of external validity of these results.

3.5 Intervention Fidelity

In this section, we provide evidence to help assuage any concerns about the implementation of the experiment. First, we show balance in baseline covariates. Then, we present information on the attrition rates. Finally, we show teachers and managers have a strong understanding of the incentive schemes. Combined, this evidence suggests the design “worked”.

Schools in the two treatment arms and control appear to be balanced along baseline covariates. Appendix Table A2 compares schools along numerous student and teacher baseline characteristics. Of 27 tests, one is statistically significant at the 10% level and one is statistically significant at the 5% level, no more than we would expect by random chance. Results presented include specifications which control for these few unbalanced variables.

Administrative data is available for all teachers and students who stay employed or enrolled during the year of the intervention. During this time 23% of teachers leave the school system, which is very similar to the historical rate of turnover. 88% of teachers completed the endline survey. While teachers were frequently reminded and encouraged to complete the survey, some chose not to. We do not see differences in these rates by treatment.

Finally, for the endline test, parents were allowed to opt out of having their children tested. Student attrition on the endline test was 13%, with 3 pp coming from students absent from school on the day of the test and the remaining 10 pp coming from parents choosing to have students opt out of the exam. On both the endline testing and endline survey, we do not find differences in attrition rate by treatment. We also do not find that lower performing students were more likely to opt out.

Teachers have a decent understanding of their treatment assignment. We asked teachers after the intervention how long it took for them to understand fully the incentive scheme they were under and the average response was 4 months. However, an important limitation of the study is that we only follow schools for one year of the incentive scheme, so teachers' effort responses may be different in the long run as they learn about how the incentive system works.

4 Results

We now present the main reduced form results of the paper. First, we test the effects of each incentive on student test performance and socio-emotional development, our two measures of student learning, V . Then, we show the effect of incentives on teacher effort, which helps to explain why we see the results on students that we do. Finally, we discuss how managers actually implemented the subjective performance arm and how that varied across managers.

4.1 Effect of Incentives on Student Outcomes

4.1.1 Specification

To test for the effect of each incentive system on student outcomes, our main specification is:

$$Y_{ij1} = \alpha + \beta_1 \textit{SubjectiveTreatment}_s + \beta_2 \textit{ObjectiveTreatment}_s + \delta Y_{ij0} + \chi_j + \epsilon_{ij} \quad (7)$$

The main dependent variable of interest is student outcome, Y_{i1} , for child, i , in subject, j , at endline, $t=1$. Student outcomes include test scores in Math, Science, English and Urdu and socio-emotional development. $\textit{SubjectiveTreatment}_s$ and $\textit{ObjectiveTreatment}_s$ are a dummy for whether the student's school, s , was assigned to subjective or objective performance raises. The left out group is the control group (flat raise). The coefficients of interest are β_1 and β_2 , and their test of equality. For test scores, we control for student's baseline score in the corresponding subject, Y_{ij0} ,

to improve efficiency as there is high auto-correlation in test scores.¹⁶ We also control for strata fixed effects, subject and grade, χ_j . Standard errors are clustered at the school level (the unit of randomization), and both standard and randomization inference p-values are provided in each table along with q-values to adjust for multi-hypothesis testing across all tests in a given table.

4.1.2 Results

Test Scores We find that both subjective and objective performance incentives have similar effects on test scores, of about 0.09 sd. Figure 2 and table 3 presents the results of each performance incentive on endline test scores. Column (1) shows results for all tests and question items. Effects are similar between the objective and subjective incentives, with an effect of 0.092 sd (p-value = 0.058) and 0.086 sd (p-value=0.022), respectively. In the row titled “F-test p-value (subj=obj)”, we present a test for the equality of $\beta_1 = \beta_2$. We cannot reject equality of effects between the two treatments on test scores. All results appear unchanged whether we consider standard p-values (in parentheses) or randomization inference p-values (in brackets), though when we correct for multi-hypothesis testing the q-value ticks up 0.156 for the effect of the objective treatment relative to control and 0.0650 for the effect of the subjective treatment relative to control. These effects appear to be fairly consistent by baseline student ability and baseline school ability. Table A3 shows that we do not find heterogeneous treatment effects by either dimension of baseline performance.

Column (2) and (3) presents treatment effects by question item type to understand whether these test score impacts are due to memorization of class content or actual learning. Column (2) only includes questions from the prior grade’s content and column (3) only includes questions that were added by the researchers from external standardized test sources including PISA, TIMSS, PERL and LEAPS.^{17,18} Both sets of questions provide evidence against the idea that incentives only improved test performance on easily “gameable” items. Remedial content (from previous grade levels) and external content had not previously been included on the school system’s standardized exam, and so teachers would not have prepared specifically for this material. Given that we find similar, if not larger effects on these types of questions, it appears that treatment effects did result in true learning gains. Again, we do not see a significant difference between the subjective or objective treatment on these items.

Column (4) and (5) present the results by subject, splitting by math and science exams versus the two reading exams (English and Urdu). Magnitudes are similar, around 0.09 - 0.10 sd, for both subjects, though the effect is no longer statistically significant for the objective treatment schools

¹⁶For grade 4 students we do not have a baseline because standardized testing in the school system begins in 4th grade. For these students, and any other that are missing a baseline, we denote a score of zero and add a dummy for having the baseline test missing.

¹⁷Not all subject and grade exams had remedial or external questions, so this is reflected in the decrease in sample size.

¹⁸Question items derived from these international sources were relevant to the curriculum of this school system and were not always matched to corresponding grade from the international exam, if that content was not part of the given year’s curriculum.

(p -value = 0.14). Again, we cannot reject equality between the two treatments and the magnitude of the effects are highly similar.

Socio-Emotional Development While the effects on test scores were similar between both treatments, the effects on socio-emotional development paint a different picture. Figure 3 and table 4 presents the results on socio-emotional development overall and broken down by socio-emotional area. Pooling across all five components of the index, we find the subjective treatment results in 0.05sd higher average score compared to the objective incentives, marginally significant at the 10% level (though not significant if we adjust for multiple hypothesis testing across all 18 tests in this table). These results come from a small negative effect of objective incentives relative to the control and a small positive effect of the subjective treatment but neither are statistically significantly different from control.

When we split these results into their sub-areas, we see that the overall negative effect of objective incentives relative to subjective is coming from a differential effect on “love of learning” and whether students like their school or would like to change schools. We can reject equality of the two treatments on these sub-areas at the 10% and 1% levels, respectively. This suggests that while objective incentives led to an increase in test scores, it was at the cost of enjoying school. Whereas, subjective incentives resulted in the same learning gains without these negative consequences. On the three other areas, ethical behavior, being a global citizen, and inquisitiveness, we cannot reject the equality of the two treatments.

We also find that students under the subjective treatment exhibit stronger resilience and sustained attention than those in the objective incentive or control schools (table 5). Question item order on the exams was randomized allowing us to use variation in performance on specific items or at specific points in the test to isolate these effects. To test for resilience we compare how students perform following an item in the top 30% of difficulty. While students in the control and objective group perform 5 p.p. worse on an item which comes immediately after a difficult question, those in the subjective treatment only perform 3 p.p. worse, and we can reject equality of the two treatments at the 10% level (table 5, col. 1).

To test for sustained attention, we compare student performance at the beginning of the test compared to items later in the exam. Figure 4 demonstrates that the pattern over the length of the test is substantially different by treatment status. Students in the objective incentive group perform better than the control in the first half of the exam but perform similarly by the end. In contrast, students in subjective incentive schools perform similarly to the control at the beginning but do substantially better in the middle and end of the exam, exhibiting less cognitive fatigue compared to the other groups. Table 5, columns 2 and 3, compares student performance by question item location in the test. We do not find a statistically significant difference if we assume a simple linear decline in performance over the test (column 2). However, if we allow a more flexible relationship, comparing performance by quintile of the test, we find that the subjective group performs significantly better

in quintiles 4-5. These results are consistent with students in the objective group working harder at the beginning of the test, perhaps because their teacher told them to take the exam seriously, but fatiguing as the test goes on. Whereas, students in the subjective schools performance is consistent with improved sustained attention (Brown et al., 2025).

4.2 Effect of Incentives on Employee Effort

4.2.1 Specification

To understand why test score effects are similar but socio-emotional development effects differ, we need to understand teacher’s behavioral response. To do this, we look at the effect of each treatment on classroom observation ratings and time use. We have a similar main specification, this time at the teacher level:

$$Y_i = \alpha + \beta_1 \textit{SubjectiveTreatment}_s + \beta_2 \textit{ObjectiveTreatment}_s + \chi_j + \epsilon_i \quad (8)$$

The main dependent variable of interest is outcome, Y_i , for teacher, i . Teacher outcomes include classroom observation scores and time use. We again control for grade and strata fixed effects, χ_j , and standard errors are clustered at the school level (the unit of randomization).¹⁹

4.2.2 Results

Classroom Observations The effect of each incentive on classroom behavior sheds light on the student effects we see. Overall, we find teachers under objective incentives using teaching strategies which provide the largest marginal return on test scores but may hamper other areas of human capital development for students. Teachers in the subjective treatment, however, do not exhibit any of those distortionary practices.

On average, objective teachers exhibit slightly worse teaching pedagogy, though the difference is not statistically significant ($p\text{-val} = 0.109$). Figure 5 and table 6 presents the effects of each incentive on teachers’ overall classroom observation score, using the CLASS rubric. Averaging across the dimensions, they score 0.07pts lower on the 7pt CLASS rubric scale (mean=4.7, sd=0.5). Subjective teachers have no noticeable change in pedagogy quality relative to the control schools.

We then break down the 12 CLASS dimensions of pedagogy into three main areas, “class climate”, “differentiation”, and how “student-centered” the lesson is. “Class climate” captures whether the atmosphere of the classroom is positive, supportive, and joyful or negative, punitive, and dull. “Differentiation” captures whether the lesson is structured in a way to meet students who have different proficiency levels and/or have different learning styles. Finally, “student-centered” measures how much of the lesson is teacher-directed versus student-involved. Teachers under the objective

¹⁹We do not control for subject here, unlike in our student specification, because most teachers teach several subjects. In addition, for classroom observations, the observation period often overlapped with several subjects.

incentive contract have a more negative class climate and less student-centered lessons. Both see a decrease of around 0.1 pts. We can reject equality of treatments at the 10% level. There is also an increase in the level of differentiation in the subjective and objective treatment schools relative to control schools.

We also measure the amount of class time devoted to test preparation activities. This includes practice tests, testing strategies (such as how to approach a multiple-choice test), or lecturing about the importance of doing well on tests. We find a large increase in the time spent on these activities in objective treatment schools. Relative to a control group mean of 0.14 min out of the 20-minute observation spent on test preparation activities, objective classes see a 5-fold increase, totaling 0.72 minutes spent on these activities. We can reject equality of treatments at the 5% level along this dimension, though the q-value to adjust for multiple-hypothesis testing is 0.105.

Together with the student outcomes, these classroom observations paint a picture of objective schools on net delivering test score gains by taking the path of least resistance for teachers – doing more test preparation and maintaining a stricter, less student-centered classroom. It is perhaps unsurprising then that we see negative effects on students’ human capital development, such as love of learning. Subjective classrooms on the other hand are able to accomplish the same academic gains without any negative effects on teacher practices or student socio-emotional development. This suggests that managers are able to prevent these distortionary behaviors, solving, at least to some extent, the multi-tasking problem.

One concern with using classroom observation data is that teachers may worry their videos or classroom observation scores will be provided to their manager, and for subjective teachers that has more of a consequence than for the other treatment arms. We do several things to help alleviate concerns there may be differential Hawthorne effects by treatment. First, in the consent form and during the camera set up, we communicate to teachers that the videos are confidential and will only be reviewed by the research team. We also let them know that only aggregated data at the school level will be provided to the school system head office. Second, visits were a surprise within a two-month window, so teachers could not adjust their lessons beforehand. Third, we recorded several hours back to back for each teacher. We find teachers are most aware of (and responsive to) the camera in the first hour of taping. As a robustness check, we restrict analysis dropping the first hour of taping and the results look, if anything, slightly stronger (table A4).

Attendance and Time at Work We find that the subjective treatment results in a significant increase in the number of days a teacher is present at work relative to no incentives. Table 7 presents the results of the biometric clock in/out data. Relative to a control group mean of 145 days, subjective teachers are present an additional 6 days. We do not find an effect on hours spent at work for either treatment relative to the control. We cannot reject equality of treatments in either outcome. Columns (2) and (4) restrict to a sample of teachers who were present in the school system both terms and did not take any long leaves (health, maternity, etc.) to ensure the days present

result is not driven by these effects. Results are robust to this sample restriction.

4.3 How do Managers Implement the Subjective Incentive?

In this section, we unpack what types of teacher actions managers value, the extent of perceived managerial bias/favoritism, heterogeneity in treatment effects by manager quality and effects of the treatments on management practices. To understand how managers use the subjective treatment arm, we draw on data from the endline teacher and manager survey, measures of teacher effort, and text of the evaluation criteria managers used to evaluate teachers. As we saw in the description of our manager sample in section 3.4, managers spend a large fraction of their time overseeing teachers, providing feedback, conducting classroom observations and generally supporting them in their development. Managers would not have access to data like a teacher's value-added, but they would have information on students' test scores each semester.

What do managers value in rating teachers? We use three approaches to help understand what types of teacher actions managers reward. In an ideal setting, we would randomize teacher actions to see how this affects managers' performance ratings of teachers. We are unable to do that exact exercise here. Instead, we use data from teacher beliefs about what actions are valuable, survey vignettes posed to managers and the correlation between teacher actions and evaluation scores. Combined, these three sources of evidence suggest that managers on net value teacher actions which are related to human capital development and are not just focused on administrative tasks or actions unrelated to student development.

Our first piece of evidence on what managers value in teachers, comes from endline survey data from both teachers and managers. We asked both teachers and managers to respond to a hypothetical situation, in which a teacher asks them for advice about how to achieve a higher raise in the following year. They are then asked to rate how much time the teacher should spend on different types of actions. Table A5 presents the data from the survey question. Column 2 shows teachers' responses about which actions would be most highly valued under the subjective contract. Column 3 presents responses to the same question posed to managers. Both subjective teacher and managers agree that improved pedagogy, like making lessons student centered and tailoring lessons to students at different initial levels, would increase their subjective rating. However, managers put additional weight on spending time collaborating with other teachers. Neither subjective teachers nor principals believe more superficial administrative tasks like volunteering at afterschool events or meeting with parents are important drivers of the subjective performance rating.

Our second piece of evidence also comes from the manager endline survey. We provide a vignette describing a hypothetical teacher to managers, and we ask them to provide a performance rating of the hypothetical teacher. The vignette randomizes the hypothetical teacher's name, and rank in

three performance characteristics: value added, classroom behavioral management and attendance.²⁰ Table A6 and figure A7 presents managers’ responses to this survey question. We find that managers highly value all three performance characteristics, but place double the weight on teacher value-added as they do on behavioral management and attendance. On average, moving from the 50th percentile value added to the 90th percentile value added would increase a teacher’s subjective rating by 0.7sd. Columns 1 through 3 of the table test each attribute separately. Column 5 adds all attributes together, and we see no difference in relative preference for these teacher characteristics when included simultaneously. These results are also robust to adding manager fixed effects (Col. 6).

Finally, we can look at what teacher behaviors are correlated with teachers’ actual performance rating in the subjective treatment arm. Table A8 and figure A10 shows the correlation between teachers’ performance rating and teacher behaviors, as measured from classroom observation data, teacher value-added and biometric clock in/out data. The subjective performance rating is positively correlated with both value added and teacher attendance. This relationship remains when we control for subject and grade (column 2) and classroom observation scores (column 3). We find mixed evidence on the relationship between pedagogy and subjective rating. Some aspects of good pedagogy are correlated with performance rating score (teachers who have a negative class climate have a lower rating) but others are not (teachers who spend more time on analysis/inquiry skills or have a more student-centered classroom are negatively rated). Two important limitations with this approach are that there are certainly omitted variables which we are unable to capture and strong co-linearity between these different aspects of pedagogy. However, combined these three pieces of evidence suggest that managers have preferences which could be moderately aligned with a reasonable social planner.

Favoritism and bias A primary concern about subjective performance pay is whether managers are biased against certain employees or show favoritism toward preferred individuals. To assess whether this is a significant concern in this setting, we ask teachers at endline whether they felt their manager discriminated against certain groups or favored certain colleagues. Table A9 presents the results from these survey questions. On average, teachers in the subjective treatment arm are no more likely than those in the objective treatment arm to say that the contract unfairly favors certain teachers or discriminated against certain groups. Teachers also state that bias, gaming and favoritism is not a significant concern in either contract. One concern with this approach is that teachers may be hesitant to provide honest assessment in a survey. To help minimize this concern teachers’ responses are anonymized and we communicate this to teachers at the time of consenting to the survey. We also ask the question several ways, including asking teachers to report such

²⁰The vignettes stated, “[Female name/Male name] is in the [bottom/middle/top] 10% of teachers in terms of students’ test score growth, in the [bottom/middle/top] 10% of teachers in terms of behavioral management, and is in the [bottom/middle/top]10% in terms of attendance and timeliness at work.” Managers rated three such vignettes with characteristics randomized across vignettes.

behavior about other schools or about the school system in general. This type of question phrasing allows teachers to report problematic manager behavior while providing plausible deniability for their own manager. We do not find a statistically significant difference in perceived bias or gaming by treatment across the various question framings.

Though teachers do not say that overt bias is a significant concern, we may be worried that there are more subtle types of bias at play. One primary type of bias we were concerned about in this setting was gender bias. In Pakistan, gender bias in employment is rampant (The World Bank Group, 2018), and, as in many settings, our managers are more likely to be male than the employees they oversee. Because understanding the existence and conditions necessary for gender bias is a substantial topic on its own, a separate paper, Brown (2025) addresses this in detail.²¹

Heterogeneity in treatment effects by manager characteristics On average the subjective treatment arm appears to have been successful at improving student outcomes and teacher effort, but there may be heterogeneity in how effectively managers implement the contract. We do not find significant differences in the effectiveness of the subjective treatment by manager gender, age or experience (table A10 col 1-3). However, we find that subjective performance pay is significantly less effective in schools where teachers believe their managers do not have an accurate perception of teacher effort. Table A10 presents heterogeneity in treatment effects by several dimensions of management “quality” (col. 4-6). We measure this by asking teachers to rate how accurate their manager is in rating a fellow teacher.²² We find there is no effect of subjective performance pay on student test scores for managers who are in the top quintile of this inaccuracy measure. This suggests that while subjective performance pay is on average very successful at producing learning gains, these contracts may be ineffective in settings where employees do not trust their managers to implement them accurately.²³

²¹To summarize the findings from that paper, managers do not appear to hold discriminatory views about the productivity or quality of teachers by gender. However, when manager evaluations affect employees’ raises, we find male teachers receive significantly higher evaluations controlling for a rich set of teacher actions and student outcomes. When evaluations are only used for feedback, but do not have a financial stake, we see no difference in evaluation score by gender. Finally, we show when managers are randomly assigned to conduct more classroom observations, this significantly reduces the extent of gender bias even when the evaluations have a financial stake. Combined, this suggests subjective evaluations can introduce systematic bias into the performance incentive system but these effects can be minimized when paired with other effective HR policies, such as better information about employee productivity or monitoring.

²²To measure whether teachers believe a given manager conducts accurate performance evaluations, we ask teachers to answer the following question about three fellow teachers in their school, “The appraisal score their manager would give them is... [Too high/low by more than one raise category], [Too high/low by about one raise category], [Too high/low by less than one raise category], or [Accurate]”. We then construct an average of these ratings per manager.

²³We do not find heterogeneity and treatment effects by world management survey overall manager score (shown in Table A10, column (5)) or personnel management sub-score (column 6). However, as discussed in section 3.3, because this data was collected from manager self-report, we should interpret the levels with caution, as managers may over rate themselves on these survey questions.

Effect of Treatments on Management Practices It is possible the treatments changed the overall culture around management and oversight in the schools. To test for this, at endline we ask teachers about the observation practices they experienced from their manager and we collect information on weekly allocation of time by managers (table A7). We find slight changes in the frequency and quality of observations relative to the control group but cannot reject equality of the subjective and Objective treatments (panel A). Treated teachers receive slightly fewer observations, perhaps because in control schools, managers believed they needed to substitute for the lack of financial incentives with more direct oversight. There does appear to be a difference in how managers spend their time across tasks in the various treatment arms (panel B). Managers spend about 6.5 hours per week meeting with teachers and staff in the control schools, compared to 7.6 hours in subjective treatment schools and 4.6 in objective treatment schools, and we can reject equality of treatments at the 1% level. This suggests that the subjective treatment also induced more overall communication between managers and their employees.

5 Mechanisms - Noise and Distortion

How can we square our main three reduced form results that we see: i). different teacher effort responses (focus on different types of pedagogy), ii). similar test score effects and iii). different socio-emotional effects across subjective and objective incentives? We argue that differences in the levels of noise and distortion across the two treatments help explain these outcomes. We structure our argument as follows.

First, in section 5.1, we present the similarities between the two treatments to help eliminate possible channels that could drive the difference in treatment effects. Second, in section 5.2, we highlight the differences between them. We show teachers believe subjective incentives to be less noisy and less distorted. Third, we provide evidence that noise and distortion does, in fact, affect outcomes. Section 5.3 shows that noise and distortion are related to student outcomes as predicted in the theoretical framework – more noise reduces the effect of incentives and more distortion diverts employee effort toward those actions. We conduct these tests by exploiting heterogeneity in levels of noise and distortion *within* a given treatment, to isolate the causal effect of noise or distortion on outcomes. Finally, in section 5.4, we bring together the estimates from section 5.2 and 5.3 to understand how much of the difference in the reduced form student effects can be explained by differences in the level of noise and distortion of the objective versus subjective treatment.

5.1 Similarities between Treatments

In order to isolate the effect of the performance metric (percentile value-added versus manager rating), we hold a number of features constant between the two treatments. Both treatments are within-school tournaments. Both treatments provide a raise from 0-10% and have the same percentile

thresholds for each raise category. Both treatments were introduced at the same time in schools and had a similar performance review timing – manager completed midterm feedback in June 2018 and final ratings in December 2018 and the objective score was based on the average of tests in June 2018 and January 2019.

At endline, we survey teachers about their experience with their incentive scheme. We find no difference in i). when teachers said they understood what was expected of them, ii). awareness of contract main features, iii). how frequently they thought about their contract, and iv). whether the system unfairly favors certain types of teachers (age, gender, etc). Figure A8 and table A9 provides results for each of these survey questions, showing no statistical difference between teachers’ responses by treatment.

5.2 Differences Across Treatments: Noise and Distortion

In this section through section 5.4, we will focus on two of the remaining differences between the treatments: the perceived noise and distortion. As highlighted in the theoretical framework, noise captures the extent to which a teacher’s actions affect their incentive payment. Distortion captures the extent to which actions with the largest marginal return to student learning also receive higher effective piece rate. First, we will show that the levels of perceived noise and distortion are different across the treatments.

Noise We measure noise using teacher’s perceptions of the noisiness of their incentive treatment.²⁴ To measure perceived noise, we ask teachers to agree or disagree (on a 5pt scale), whether under their contract, “their raise is out of their control”, “those who work harder, earn more” and whether “I feel motivated to work harder”. Figure 6 presents the average response to each question with 1 being strongly disagree and 5 being strongly agree and table 8 presents results converting responses to z-scores. Teachers in the subjective treatment feel their raise is more within their control, believe hard work is rewarded, and report feeling more motivated. The average difference is 0.145 sd pooling across the three questions, and we can reject equality of treatments at the 1% level (table 8).

Teachers appear to be relatively correct in their assessment of which contract is noisier. To test the “actual” noise, we compare the R^2 , from a “kitchen sink” regression, where we regress the raise they received on everything we can see about the teacher’s actions (16 dimensions of classroom observation scores, time use, and attendance). Using a likelihood ratio test, we find that teacher actions are substantially more predictive of subjective performance evaluation scores than objective evaluation scores ($p < 0.001$). This suggests that teacher actions are more related to their subjective score than their value-added, though we cannot determine if this is causal.

²⁴We think this is preferred to using “actual” noise, measured by seeing how predictive teacher’s measured behavior is to their raise. Perceived noise is what matters for teacher’s behaviors this last year, and there is likely measurement error that is correlated with treatment in measuring “actual” noise.

Distortion In order to understand the extent of distortion, we need to know what teachers believe is the “effective piece rate” (g) to different actions. At endline, we ask teachers to imagine a teacher who really wants to receive a higher raise at the end of the year and commits to work ten additional hours a week to increase their raise. Then we ask how those ten hours should be allocated across different activities, such as collaborating with other teachers, incorporating higher order thinking skills into lessons, preparing practice tests, helping with extracurricular activities, etc.

We find that teachers in subjective versus objective schools feel that there are some differences in which actions should be prioritized in order to increase their raise. Figure 7 and table A5 shows the difference in beliefs across all 11 actions. Table 9 presents the differences in stated valuation pooling the 11 actions into four main categories: administrative tasks (grading, helping with extracurriculars, monitoring duty, etc.), professional development (collaboration, training, content knowledge, etc.), pedagogy (use of student-centered and differentiated lessons), and test preparation (achieving certain grade targets).

Overall, teachers think those under the subjective contract should prioritize making lessons more student-centered and focus less on behavioral management. We will show in the next section that focusing more on these actions have different implications for student outcomes.

5.3 Effect of Noise and Distortion on Student Outcomes

Noise We showed that teachers believe there is less noise in the subjective performance measure. However, we do not know if noisier incentives produce a smaller effort response. We showed that theoretically with a fixed variance incentive scheme, a more noisy incentive scheme leads to a lower powered incentive, but there is no empirical evidence on this effect.

To test whether noise affects outcomes, we exploit heterogeneity *within* the subjective treatment in noisiness. Managers vary in their accuracy of assessing teacher quality. Some managers observe lessons for each of their teachers every week, others sit down and review paper lesson plans, and some are completely hands off. To measure whether a manager is perceived to give accurate performance evaluation scores, we ask teachers to answer the following question about three fellow teachers in their school who are under the same manager as the surveyed teacher, “The appraisal score their manager would give them is... [Too high/low by more than one raise category], [Too high/low by about one raise category], [Too high/low by less than one raise category], [Accurate], ...”. We then construct an average of these ratings per manager, capturing average *perceived* manager inaccuracy. On average, teachers believe their managers over or under rate their fellow teachers by 0.8 of an appraisal step (out of the five-step system shown in section 3.2). However, there is considerable heterogeneity—managers in the most inaccurate quintile are perceived to over/under rate other teachers by greater than two steps (figure A9).

Of course, schools with more inaccurate managers are likely different in many ways (experience of manager, student composition, etc). However, this measure should only affect teacher’s perceived

noisiness of the incentive scheme in subjective treatment schools. In control or objective treatment schools, managers still rate their teachers but have no control over the incentive raise in those schools. Therefore, we use $ManagerInaccuracy * SubjectiveTreatment$ as the instrument for $Noise$ of the contract, controlling for $ManagerInaccuracy$ and $SubjectiveTreatment$.

We find that $ManagerInaccuracy_j$ significantly increases teacher’s rating of the noisiness of their appraisal system in subjective relative to control/objective treatment schools. A 1 sd increase in manager inaccuracy increases beliefs about the noisiness of the contract by 0.1-0.4 sd in subjective schools. Table 10 presents the results from the first stage for data at the teacher and student level.²⁵ Columns (2) and (4) add additional controls, including teacher’s beliefs about the preference for different actions (“distortion”) and teacher beliefs about other non-noise features of the contract (timing, understanding, etc) in levels and interacted with treatment. The coefficient on $ManagerInaccuracy * SubjectiveTreatment$ is very robust to the inclusion of these controls, suggesting that this instrument is picking up difference in noise and not other features of the contract environment.

To test for the effect of noise on teacher and student outcomes, we use the following two-stage least squares specification:

$$Outcome_{ij} = \alpha_0 + \alpha_1 ManagerInaccuracy_j + \alpha_2 SubjectiveTreat_i + \alpha_3 \widehat{Noise} + \chi_{ij} + \epsilon_{ij} \quad (9)$$

where α_3 is the coefficient of interest, $Noise$ is instrumented using $Manager Inaccuracy_j * SubjectiveTreat_i$. χ_{ij} are controls, such as school and grade and baseline controls when available for a given outcome.

We find that noise significantly reduces the effectiveness of performance incentives (table 11). A 1 sd increase in noisiness of the incentive scheme reduces teachers’ hours worked by 13.2 hours per week and reduces test scores by 0.175 sd. Because our effective first stage has an f-stat of less than 10, we present the AR test p-values, our preferred test given that they are robust to weak instruments in the just-identified case. To benchmark these estimates, the perceived difference in the level of noise between the subjective and objective arm is 0.145sd, so for a noise difference of that much it would correspond to a difference in test scores of 0.02sd and a difference in hours worked of 1.9 hours.

The main exclusion restriction violation we would be worried about is that inaccurate managers may implement the subjective contract differently in other ways. For example, maybe they prioritize different evaluation criteria or reminded teachers about the contract less frequently. Columns (2), (4), and (6) add in the same additional controls as in table 10 for non-noise features of the contract

²⁵The first stage in table 10 columns (1) and (2) is at the level of the teacher used for the hours results in table 11, column (1) and (2). The first stage in columns (3) and (4) is at the level of the student and used for the student test and socio-emotional skills outcomes in table 11, column (3)-(6).

environment and their interaction with the treatment variable. The effect of noise on hours worked and test scores is robust to the addition of those controls. Our first stage is extremely weak for the effect on socio-emotional scores, and we do not find an overall second stage effect.

To further explore these possible exclusion restriction violations, we can also explicitly see whether our instrument is correlated with other outcomes (figure A11). As we would expect, managers who are believed to be inaccurate in rating their employees are different. They conduct fewer observations and are perceived to be biased toward their friends and more experienced teachers. However, importantly, our instrument $Manager\ Inaccuracy_j * SubjectiveTreat_i$ is not correlated with these outcomes. Instead, we see that $Manager\ Inaccuracy_j * SubjectiveTreat_i$ is correlated with teacher’s beliefs about the performance appraisal system. Those with inaccurate managers who are in the subjective treatment are much more likely to say the appraisal system is unfair and that they dislike it overall.

Distortion Distortion is a measure of the correlation between the marginal returns of different actions on student learning and the effective piece rates of those actions. To measure distortion we need an estimate of marginal returns to different actions. To calculate these marginal returns, we again exploit heterogeneity across managers interacted with treatment status.

The idea behind this strategy is that managers have different preferences for actions – some state they want teachers to focus more on improving their lesson plans, others want teachers to help out more with administrative tasks, etc. We can measure those preferences for all managers because irrespective of treatment status managers delineate the performance evaluation criteria in which they will rate teachers. However, it is only in the subjective schools that there is a financial stake of these evaluation criteria. We interact the weight placed on a given action category with subjective treatment status versus objective and control. In effect, we are comparing two teachers whose managers put identical value on a given action, e.g. time on test prep, but one teacher is in a subjective treatment school and one is in an objective or control school. Therefore, we expect the one in the subjective school to increase their effort more toward that action because there is a financial incentive, giving us exogenous variation in teacher effort on that action.

To test for the effect of additional effort in a given action on student outcomes, we estimate:

$$\begin{aligned}
 StudentOutcome_i = & \alpha_0 + \alpha_1 SubjectiveTreat_i + \sum_{j=1}^5 \delta_j Points\ on\ Action\ j_i \\
 & + \sum_{j=1}^5 \beta_j Points\ on\ Action\ j_i * SubjectiveTreat_i + \chi_{ij} + \epsilon_i
 \end{aligned} \tag{10}$$

The coefficient of interest, β_j , captures the effect on student outcomes of having a financial stake on action j . *Points on Action j* is not correlated in levels or in the interaction with subjective treatment to the number of classroom observations teachers receive or beliefs about manager bias

(fig. A12). We do find slight differences in beliefs about whether the appraisal system induces competition by *Points on Action j*, but we control for this in eq. 10 along with other features of the contract environment, such as noisiness, understanding of the contract, etc.

We find that having financial stakes on a given action category is related to student outcomes for several action-outcome pairs. Table 12 presents the β_j 's for each action category. Professional development and test prep actions are positively related to student test scores. However, test prep is negatively related to student socio-emotional scores. These results are robust to the inclusion of additional controls about the contract environment (table 12, col. 2 and 4).

5.4 Contribution of Noise and Distortion to Reduced Form Effects

Finally, we pull the results together in Figure 8 to understand the extent noise and distortion explain the reduced form results in section 4.1. To do this we decompose the total reduced form effect into the components from noise, distortion and an unexplained component, ϵ :

$$dTestScore = \frac{\partial TestScore}{\partial Noise} * dNoise + \frac{\partial TestScore}{\partial Distortion} * dDistortion + \epsilon \quad (11)$$

We use the following statistics to estimate each term in eq. 11. $dTestScore$ is the difference in overall effect on test scores which is -0.006 sd (table 3, col. 1). $\frac{\partial TestScore}{\partial Noise}$ is the marginal effect of noise on test scores which is -0.175 (table 11, col. 3). For $dNoise$, we use teachers beliefs about the perceived difference in noisiness. Teachers perceive the subjective arm is 0.145sd less noisy than the objective arm (figure 8, col. 1). For the distortion component, we repeat the same approach for each of the four action categories (admin, professional development, pedagogy and test prep) and sum across. Our estimate of $\frac{\partial TestScore}{\partial Distortion}$ comes from the return to different actions shown in table 12, col. 1. Our measure of $dDistortion$ is the perceived value of certain action shown in table 9.

In summary, the decomposition then is:

$$\begin{aligned} dTestScore &= \alpha_3 * dPerceived\ Noisiness + \sum_{j=1}^4 \beta_j * Perceived\ Return_j + \epsilon \\ -0.006sd &= -0.175 * -0.145sd \quad + \quad -0.0144sd \quad + \epsilon \\ \epsilon &= -0.0169sd \end{aligned}$$

Combined, the positive effect of subjective having less noise and the negative effect of teachers placing less focus on test scores under subjective somewhat cancel each other out. Overall the noise and distortion channels predict that under the subjective arm, test scores should actually be slightly higher than in the objective arm when in reality it is the reverse, though the difference is small.

We can repeat the same approach for socio-emotional skills.

$$\begin{aligned}
dSEScore &= \frac{\partial SEScore}{\partial Noise} * dNoise + \frac{\partial SEScore}{\partial Distortion} * dDistortion + \epsilon \\
0.051sd &= -0.0326 * -0.145sd + 0.0151sd + \epsilon \\
\epsilon &= 0.031sd
\end{aligned} \tag{12}$$

The overall effect of subjective relative to objective on socio-emotional development was 0.0504sd (table 4). The effect of noise on socio-emotional skills is -0.0326sd, and noise is 0.145sd lower in the subjective arm relative to the objective arm. Teachers in the subjective arm focus more on actions which are related to socio-emotional skills. Overall $\frac{\partial SEScore}{\partial Distortion} * dDistortion$ is 0.0151 sd. The remaining unexplained portion is 0.031sd. In this case, the noise and distortion channels explain a decent portion of the difference in effect between the two treatment arms.

6 Conclusion

In this paper, we provide evidence on the effect of subjective versus objective incentives for teachers. We find that both subjective and objective incentives increase test scores, but objective incentives result in negative effects on socio-emotional development. These student outcomes align with the teacher behaviors observed under each incentive. In subjective treatment schools, teachers make small improvements in pedagogy and are involved in more professional development. In objective treatment schools, teachers distort effort toward test preparation. They spend much more time on practice tests and test strategies while using more punitive discipline. While there is heterogeneity in manager application of the subjective treatment arm, we do not find evidence of widespread favoritism or bias.

We then try to understand the mechanisms underlying the reduced form effects. We show evidence that the two incentive schemes are similar along most dimensions except for two areas: noise and distortion. We show teachers believe that the subjective incentive is less noisy and that it prioritizes both test and non-test student outcomes. Using heterogeneity within treatments we attempt to isolate the effect of noise and distortion itself on student outcomes. Finally, we show that noise and distortion are able to explain a large portion of the reduced form test score effects but a smaller fraction of the reduced form socio-emotional skill effect.

The broader implications of these results are that accountability systems for teachers or school systems which rely on test scores could have detrimental effects on more well-rounded aspects of student human capital. This could be exacerbated when the same performance metric (i.e. test scores) is used both as the performance metric to reward teachers or schools and as the gauge of whether the performance incentive accountability system as a whole was effective at increasing overall system performance. Many school systems as a result of these concerns have instead developed

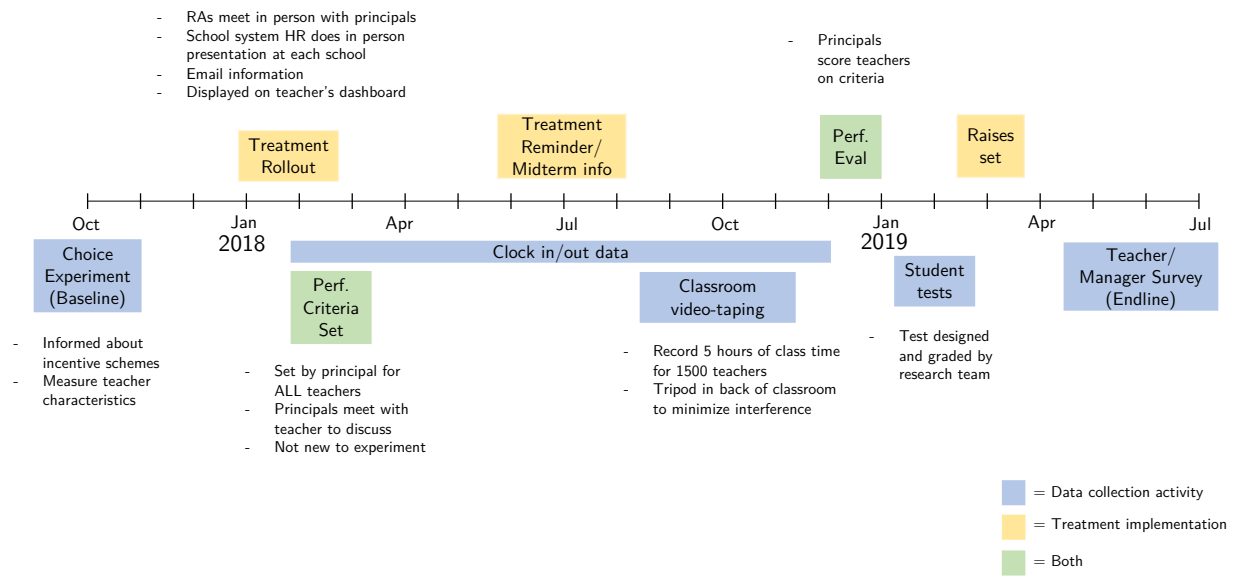
extremely complicated formulas combining numerous aspects of objective performance. However, previous research has found these systems are often ineffective. Our results show this may be due to the increased noise of these systems when the formulas are difficult for employees to parse.

Finally, despite managers having imperfect information about employee performance they appear to be effective at motivating additional effort and are able to ensure teachers do not distort their effort toward only one aspect of student development. However, this may not always be the case in all organization settings and future work could explore under what conditions managers are likely to be more effective than objective incentives and vice versa.

We emphasize that our results demonstrate the importance of the noise and distortion framework for understanding the relative effectiveness of different incentive types, rather than claiming that subjective incentives will always dominate objective ones. The relative noise of each incentive type is context-dependent. In our setting, the objective metric — simple pre/post percentile value-added within a single year — is relatively noisy compared to more sophisticated measures such as the multi-year teacher value-added methodology of Chetty, Friedman, and Rockoff (2014), which filters out transitory classroom shocks. Similarly, the subjective evaluations in our setting benefit from principals who conduct nearly twice as many classroom observations as US principals. In other labor market contexts, such as commission-based sales where output is easily measured, objective metrics may be less noisy than subjective evaluations. The broader implication of our results is that the relative effectiveness of incentive types depends on their relative noise and distortion — features that are setting-specific — rather than that subjective incentives universally dominate objective ones.

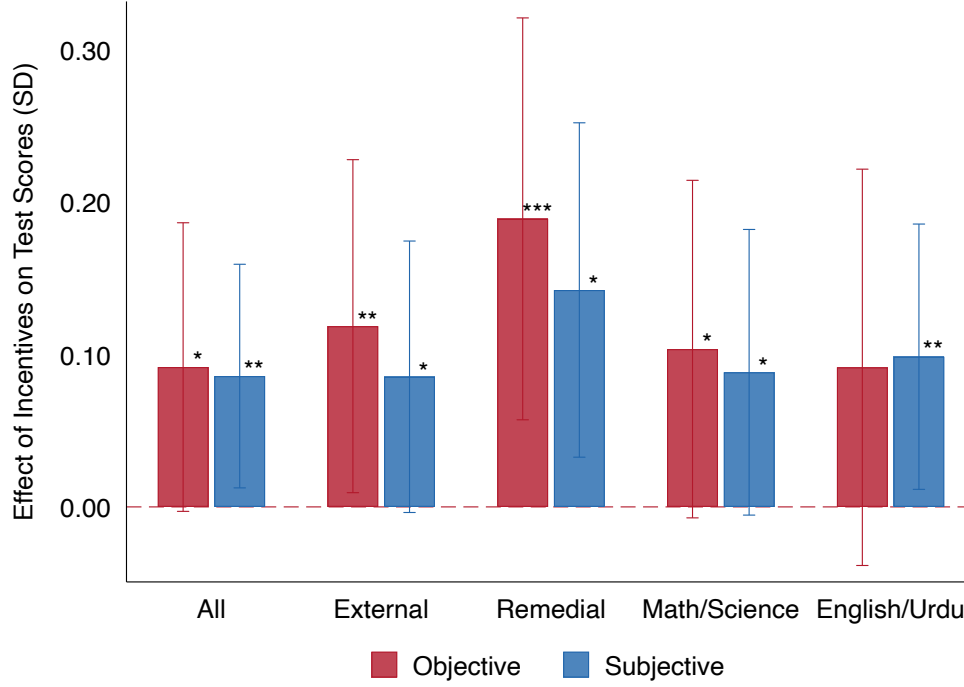
7 Figures

Figure 1: Experimental Timeline



Notes: This figure presents the experimental timeline. It includes data collection activities and treatment implementation activities.

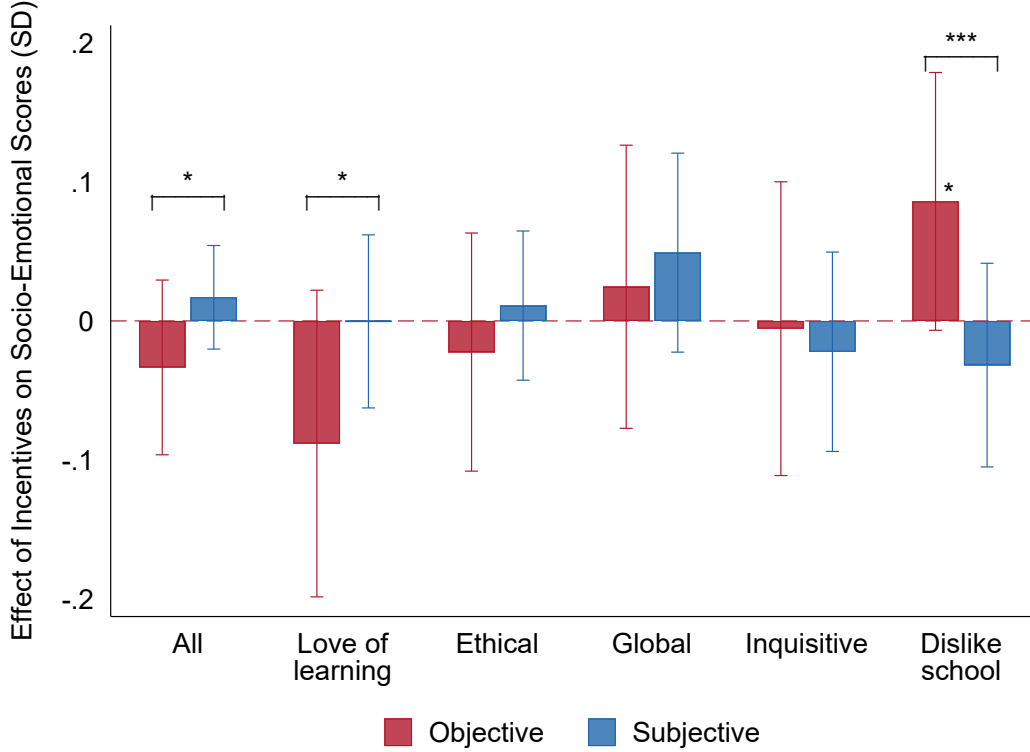
Figure 2: Effect of Incentives on Student Test Scores



Notes: This figure presents the effects of each performance incentive treatment on student endline test scores relative to the control group.

- The red bars present the coefficient of the effect of the objective treatment relative to the control (flat raises). The blue bars present the coefficient of the effect of the subjective treatment relative to the control.
- The observation is at the student-subject level. The y-axis presents the coefficient from a regression of student's z-score on a given endline exam on treatment dummy variables.
- The sample includes students tested in grades 4-13 in five subjects: Math, Science, English, Urdu, Economics.
- The first two bar graphs includes all test subjects and question items. The next two bars restricts to math and science exams. The next restrict to English, Urdu and Economics exams. The next restrict to question items drawn from external sources, such as PISA and TIMSS. The last two restrict to question items which were from the previous grade.
- All regressions include strata fixed effects and control for baseline student average test score, baseline school average test score, grade and subject. Standard errors are clustered at the school level. 95% confidence intervals are shown on each bar. Stars just above a bar show the significance of the treatment group relative to the control. A bracket above two bars denotes the significance between the two treatments (subjective versus objective). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

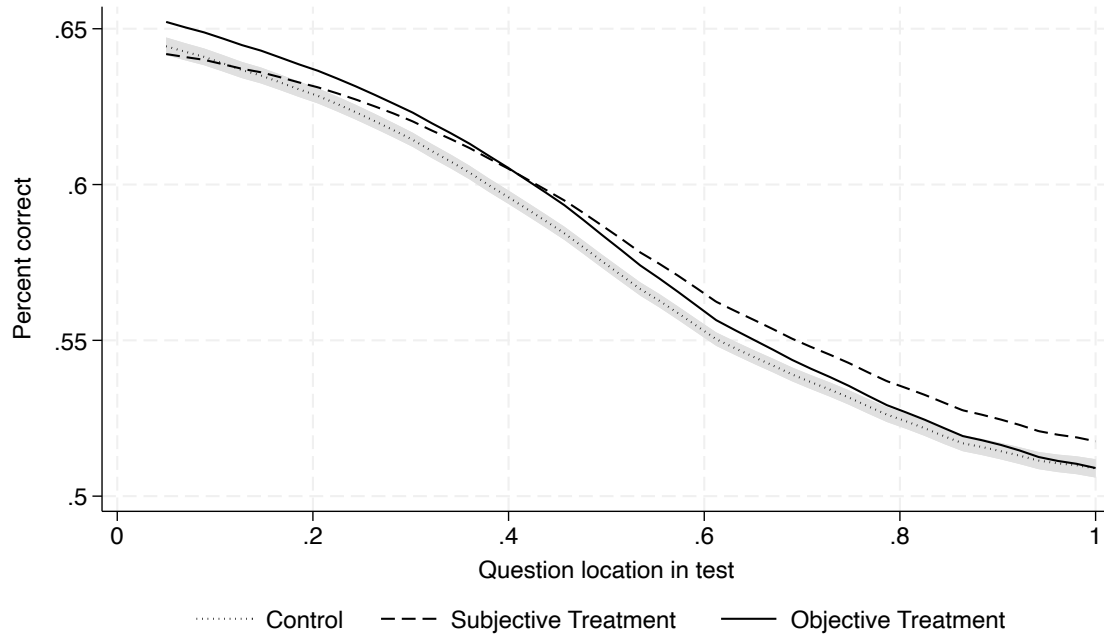
Figure 3: Effect of Incentives on Socio-Emotional Outcomes



Notes: This figure presents the effects of each performance incentive treatment on student socio-emotional outcomes relative to the control group.

- The red bars present the coefficient of the effect of the objective treatment relative to the control (flat raises). The blue bars present the coefficient of the effect of the subjective treatment relative to the control.
- The observation is at the student level. The y-axis presents the coefficient on a regression of student's z-score on a given socio-emotional dimension from an endline survey of students conducted in January 2019.
- The first two bars provides the average across all five dimensions of socio-emotional outcomes. The remaining ten provide effects on each individual dimension.
- All regressions include strata fixed effects and control for student's grade. Standard errors are clustered at the school level. 95% confidence intervals are shown on each bar. Stars just above a bar show the significance of the treatment group relative to the control. A bracket above two bars denotes the significance between the two treatments (subjective versus objective). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

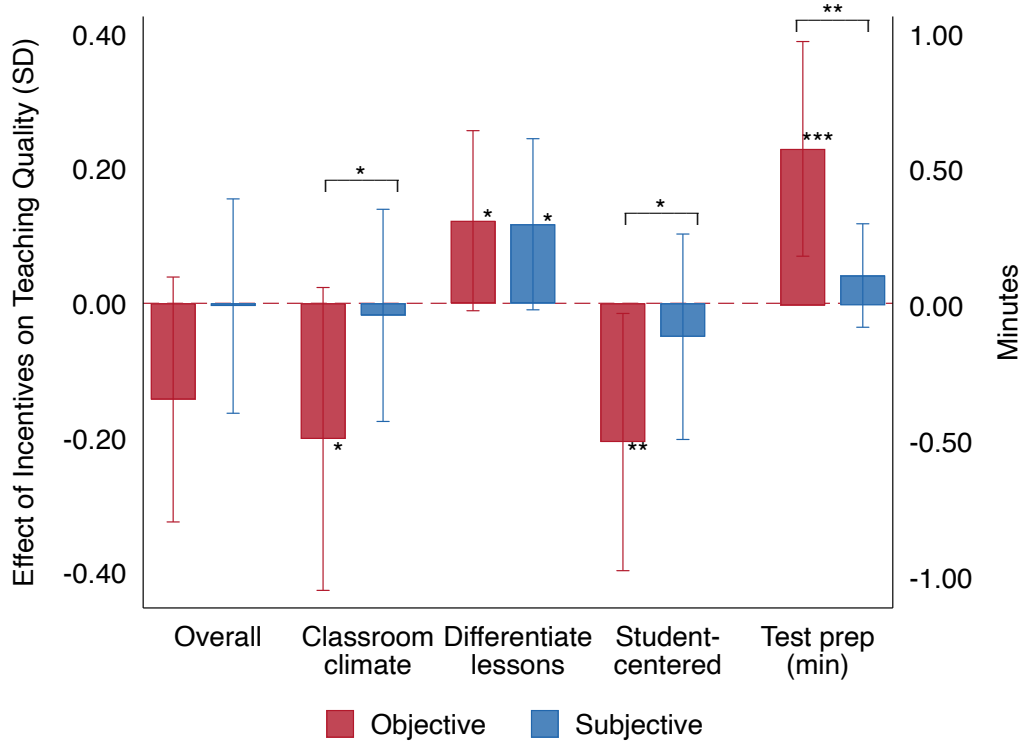
Figure 4: Effect of Incentives on Student Sustained Attention



Notes: This figure plots a kernel weighted local polynomial smoothed function of performance over the length of each endline test by treatment group.

- The observation is at the student-question item level. The y-axis presents the percent of question items correctly answered. The x-axis presents the location of the question item in the test.
- The sample includes students tested in grades 4-13 in five subjects: Math, Science, English, Urdu, Economics.
- The dotted line shows the performance of the control (flat raises) with 95% confidence intervals. The dashed and solid line show the performance for the subjective and objective treatment group, respectively.

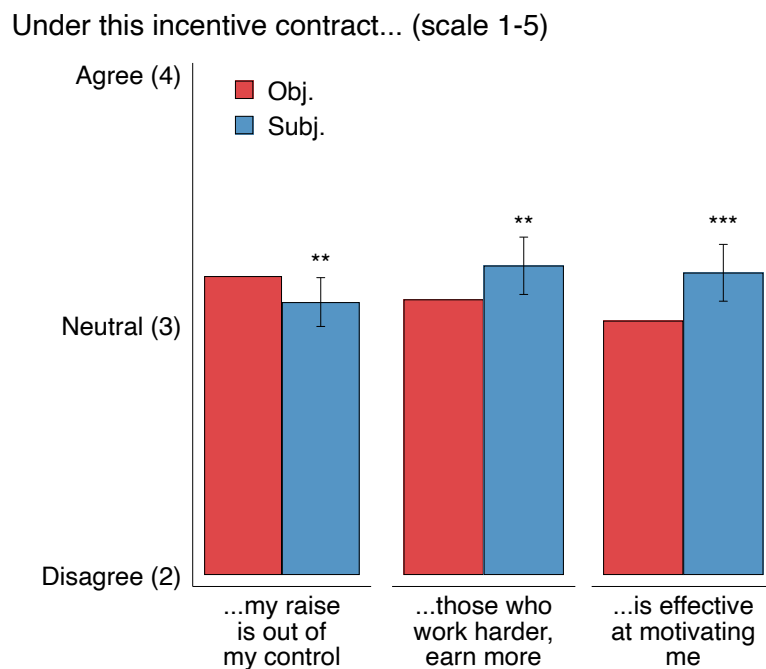
Figure 5: Effect of Incentives on Teacher Pedagogy



Notes: This figure presents the effects of each performance incentive treatment on teacher behavior as rated based on classroom videos relative to the control group.

- The red bars present the coefficient of the effect of the objective treatment relative to the control (flat raises). The blue bars present the coefficient of the effect of the subjective treatment relative to the control.
- The observation is at the classroom observation level. Teachers may be observed multiple times over the course of the intervention. The y-axis presents the coefficient from a regression of classroom observation score in a given dimension on treatment dummy variables.
- The sample includes teachers from grades 4-13 in core academic subjects.
- The first two bars presents the effects on the average score on the CLASS rubric (Pianta et al., 2012), on a 7-pt scale. The next six bars provide effects on scores on sub-areas of the class rubric. The last two bars provide effects on time spent on testing or test-prep activities (in minutes).
- All regressions include strata fixed effects and control for grade and video coder fixed effects. Standard errors are clustered at the school level. 95% confidence intervals are shown on each bar. Stars just above a bar show the significance of the treatment group relative to the control. A bracket above two bars denotes the significance between the two treatments (subjective versus objective). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 6: Difference in Noise by Treatment

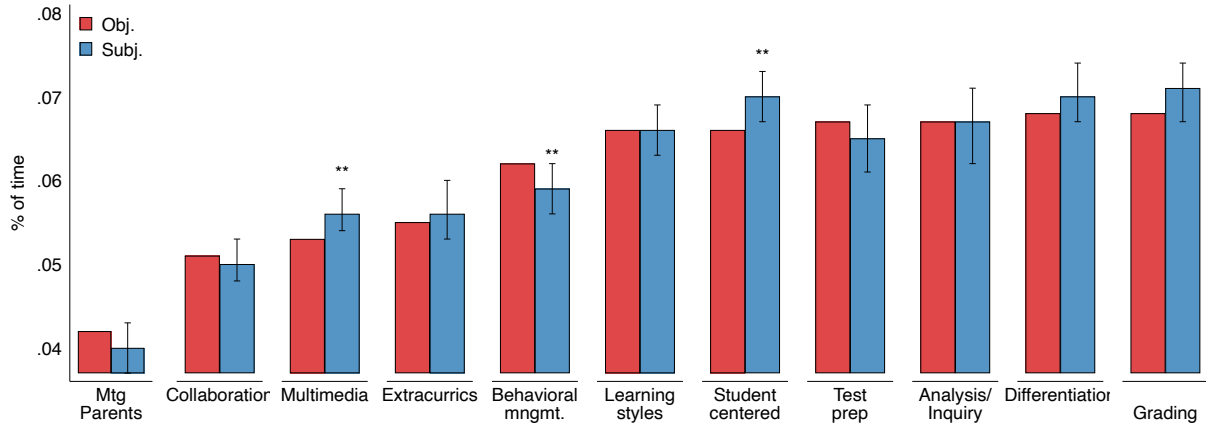


Notes: This figure presents teacher's responses to a questions about how they respond to their incentive.

- The red (blue) bars presents the average response for teachers in the objective (subjective) treatment schools. The observation is at the teacher level and come from the endline survey of teachers.
- The questions are on a 5pt scale from Strongly Disagree (1) to Strongly Agree (5).
- Standard errors are clustered at the school level. 95% confidence intervals are shown on the subjective bar comparing it to the objective treatment. Stars just above the bar show the significance of the subjective group relative to the objective. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 7: Difference in Value of Teacher Activity by Treatment

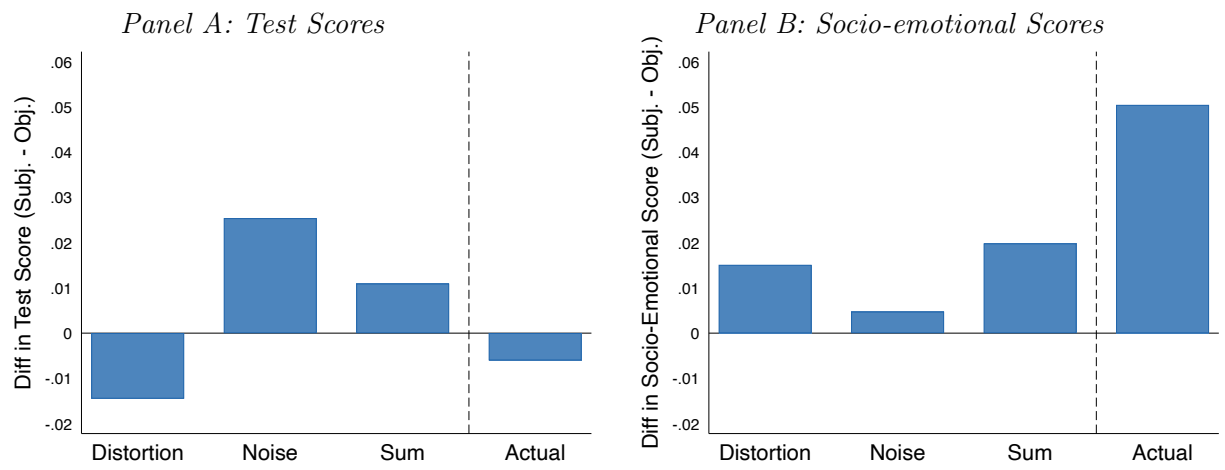
To increase my raise I should spend time on...



Notes: This figure presents teachers' responses to a hypothetical scenario in which they are advising a teacher which actions they should take to increase their raise under a given treatment given a fixed 10 hours to split across different actions.

- The red (blue) bars presents the average response for teachers in the objective (subjective) treatment schools. The observation is at the teacher level and come from the endline survey of teachers.
- The outcome is the percent of time out of the 10 total hours they could allocate across all actions.
- Standard errors are clustered at the school level. 95% confidence intervals are shown on the subjective bar comparing it to the objective treatment. Stars just above the bar show the significance of the subjective group relative to the objective. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 8: Decomposing Total Effect into Noise and Distortion Components



Notes: This figure shows the effect from noise and distortion channels compared to the total effect of the subjective contract on test scores (Panel A) and socio-emotional scores (Panel B).

8 Tables

Table 1: Descriptive Statistics about Teachers in Study and Comparison Sample

	Study Sample		US Sample	
	Mean	St. Dev.	Mean	St. Dev.
	(1)	(2)	(3)	(4)
<i>Panel A. Teacher Characteristics</i>				
Age	35.0	8.9	41.8	7.5
Female	0.80	0.40	0.76	0.43
Years of experience	5.1	5.2	13.8	9.6
Has Post BA Education	0.68	0.47	0.54	0.50
Salary, USD(PPP)	17,160	5,700	52,400	18,400
<i>Panel B. Teacher Evaluation</i>				
Number of observations per year	4.7	8.2	2.5	2.9
Use evaluation for compensation	-	-	0.12	0.32
Frequency of evaluation (months)	-	-	13.0	7.0
Performance metric used for evaluation:				
- Principal evaluation	-	-	0.90	0.30
- Test scores	-	-	0.35	0.48
- Peer evaluations	-	-	0.26	0.44
- Student ratings	-	-	0.05	0.22

Notes: This table reports summary statistics on teacher characteristics, monitoring and evaluation for our study sample and a comparison sample of teachers in US schools. Data in panel A, columns (1) and (2) comes from administrative data collected from our partner school system. Data in panel B, columns (1) and (2) is from an endline survey conducted with 5,698 teachers in our study sample. Data in panel A and B,, columns (3) and (4) comes from 42,020 teachers surveyed in the *School and Staffing Survey* (National Center for Education Statistics, 2011). Most of panel B is not included for our sample as the experiment determined these features.

Table 2: Descriptive Statistics about Managers in Study and Comparison Sample

	Study Sample		US Sample	
	Mean	St. Dev.	Mean	St. Dev.
	(1)	(2)	(3)	(4)
<i>Panel A. Manager Characteristics</i>				
Age	44.9	9.2	48.8	9.7
Female	0.61	0.49	0.53	0.50
Years of experience	9.6	7.9	13.0	7.5
Salary, USD(PPP)	45,400	34,400	85,400	29,400
<i>Panel B. Manager Time Use</i>				
Total hours worked	47.2	16.3	57.0	13.2
Hours spent on:				
- Administrative tasks	18.5	10.3	18.2	2.3
- Teacher management and teaching	17.5	8.2	15.1	2.0
- Student and parent interactions	6.3	4.4	20.2	2.7
- Other tasks	6.9	12.3	4.0	2.6
<i>Panel C. Management Practice Rating</i>				
Overall Management Score (out of 5)	4.27	0.43	2.76	0.43
People management (out of 5)	4.14	0.53	2.51	0.49
Operations (out of 5)	4.32	0.61	2.89	0.49
Performance monitoring (out of 5)	4.32	0.49	2.81	0.75

Notes: This table reports summary statistics on manager characteristics, time use and management practices for our study sample and a comparison sample of managers in US schools. Data in panel A, columns (1) and (2) comes from administrative data collected from our partner school system. Data in panel B and C, columns (1) and (2) is from an endline survey conducted with 189 principals and vice principals in our study sample. Data in panel A and B, columns (3) and (4) comes from 9235 principals surveyed in the *School and Staffing Survey* (National Center for Education Statistics, 2011). Data in panel C, columns (3) and (4) is from the *World Management Survey* data conducted by the Centre for Economic Performance (Bloom et al., 2015). We restrict to the 270 schools located in the US from that sample.

Table 3: Effect of Incentives on Student Test Scores

	Endline Test (z-score)				
	All b/p/ri/qi	Remedial b/p/ri/qi	External b/p/ri/qi	Math/Science b/p/ri/qi	English/Urdu b/p/ri/qi
Objective Treatment	0.0918* (0.0575) [0.0730] {0.156}	0.189*** (0.00518) [0.0260] {0.0650}	0.119** (0.0335) [0.0200] {0.0650}	0.104* (0.0668) [0.194] {0.291}	0.0917 (0.166) [0.144] {0.240}
Subjective Treatment	0.0859** (0.0220) [0.0130] {0.0650}	0.142** (0.0113) [0.0240] {0.0650}	0.0855* (0.0601) [0.0170] {0.0650}	0.0884* (0.0646) [0.121] {0.227}	0.0986** (0.0267) [0.0260] {0.0650}
F-test pval (subj=obj)	0.895	0.380	0.431	0.773	0.895
Randomiz infer pval (subj=obj)	0.884	0.453	0.388	0.819	0.873
FDR q-val (subj=obj)	0.884	0.566	0.529	0.884	0.884
Control Group Mean	-0.04	-0.09	-0.05	-0.04	-0.04
Clusters	234	204	225	223	225
Observations	141566	31944	100318	72714	68852

Notes: This table presents the effects of each performance incentive treatment on student endline test scores. The outcome is student's z-score on a given endline exam. The sample includes students tested in grades 4-13 in five subjects: Math, Science, English, Urdu, Economics. Column (1) includes all test subjects and question items. The observation is at the student-subject exam level. Column (2) restricts to question items which were from the previous grade. Column (3) restricts to question items drawn from external sources, such as PISA and TIMSS. Column (4) restricts to math and science exams. Column (5) restricts to English, Urdu and Economics exams. All regressions include strata fixed effects and control for baseline student average test score, baseline school average test score, grade and subject. Values in parentheses are standard p-values. Values in brackets are randomization inference p-values. Standard errors are clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Effect of Incentives on Student Socio-Emotional Outcomes

	Endline Survey Indices (z-score)					
	All b/p/ri/qi	Love of learning b/p/ri/qi	Ethical b/p/ri/qi	Global b/p/ri/qi	Inquisitive b/p/ri/qi	Dislike school b/p/ri/qi
Objective Treatment	-0.0334 (0.294) [0.395] {0.857}	-0.0882 (0.116) [0.0970] {0.437}	-0.0224 (0.605) [0.670] {0.857}	0.0246 (0.634) [0.670] {0.857}	-0.00556 (0.917) [0.930] {0.985}	0.0860* (0.0688) [0.129] {0.464}
Subjective Treatment	0.0170 (0.368) [0.560] {0.857}	-0.000335 (0.992) [0.993] {0.993}	0.0110 (0.685) [0.762] {0.857}	0.0491 (0.177) [0.228] {0.684}	-0.0222 (0.542) [0.629] {0.857}	-0.0318 (0.392) [0.451] {0.857}
F-test pval (subj=obj)	0.094	0.079	0.402	0.579	0.724	0.001
Randomiz infer pval (subj=obj)	0.094	0.025	0.511	0.609	0.726	0.008
FDR q-val (subj=obj)	0.437	0.225	0.857	0.857	0.857	0.144
Control Group Mean	-0.004	-0.002	-0.002	0.001	-0.008	0.376
Clusters	126	126	126	125	126	124
Observations	15418	15401	14904	14168	14909	11505

Notes: This table presents the effects of each performance incentive treatment on student socio-emotional outcomes. The outcome is student’s z-score on a given socio-emotional dimension. Observations are at the student level and come from an endline survey of students in January 2019. Column (1) provides the average across all five dimensions of socio-emotional outcomes after reversing the sign of “Dislike school”. Columns (2)-(6) provide each individual dimension. All regressions include strata fixed effects and control for student’s grade. Values in parentheses are standard p-values. Values in brackets are randomization inference p-values. Standard errors are clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Effect of Incentives on Student Sustained Attention and Resilience

	Question Correct		
	(1)	(2)	(3)
Subjective Treatment	0.00995 (0.00733)	0.00503 (0.0131)	-0.00700 (0.0102)
Objective Treatment	0.0165* (0.00967)	0.0237 (0.0182)	0.0149 (0.0126)
After difficult item	-0.0519*** (0.0102)		
Subjective Treatment* After difficult item	0.0230** (0.0108)		
Objective Treatment* After difficult item	0.00627 (0.0127)		
Location (percent of test)		-0.211*** (0.0179)	
Subjective Treatment* Location (percent of test)		0.0190 (0.0190)	
Objective Treatment* Location (percent of test)		-0.0108 (0.0304)	
Location Quintile 2-3			-0.0833*** (0.0160)
Location Quintile 4-5			-0.171*** (0.00667)
Subjective Treatment* Location Quintile 2-3			0.0288* (0.0164)
Objective Treatment* Location Quintile 2-3			0.00802 (0.0201)
Subjective Treatment* Location Quintile 4-5			0.0258*** (0.00771)
Objective Treatment* Location Quintile 4-5			-0.000488 (0.0144)
P-value (interaction*subj=interaction*obj)	0.051	0.240	0.101
P-value (interaction2*subj=interaction2*obj)			0.050
Control Group Mean	0.573	0.573	0.573
Clusters	234	234	234
Observations	2056337	2056337	2056337
Controls			
Baseline	X	X	X
Grade/Subject	X	X	X

Notes: This table presents the effects of each performance incentive treatment on student endline test items, comparing performance on question items just after a difficult item (Col 1) or at different locations in the test (Col 2 & 3). The outcome is whether a student got a given question correct. The sample includes students tested in grades 4-13 in five subjects: Math, Science, English, Urdu, Economics. The observation is at the student-question item level. All regressions include strata fixed effects and control for baseline student average test score, baseline school average test score, grade and subject. Standard errors are clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Effect of Incentives on Teacher Pedagogy

	CLASS Rubric				Test Prep
	All b/p/ri/qi	Class Climate b/p/ri/qi	Differentiation b/p/ri/qi	Student-Centered b/p/ri/qi	Minutes b/p/ri/qi
Objective Treatment	-0.0713 (0.123) [0.171] {0.257}	-0.0791* (0.0788) [0.101] {0.204}	0.110* (0.0719) [0.149] {0.248}	-0.115** (0.0346) [0.0480] {0.204}	0.577*** (0.00455) [0.0120] {0.105}
Subjective Treatment	-0.00206 (0.959) [0.946] {0.946}	-0.00704 (0.822) [0.838] {0.946}	0.105* (0.0699) [0.0690] {0.204}	-0.0276 (0.521) [0.559] {0.762}	0.110 (0.255) [0.649] {0.811}
F-test pval (subj=obj)	0.103	0.100	0.926	0.091	0.017
Randomiz infer pval (subj=obj)	0.109	0.083	0.940	0.094	0.014
FDR q-val (subj=obj)	0.204	0.204	0.946	0.204	0.105
Control Group Mean	4.67	5.64	2.65	4.93	0.14
Control Group SD	0.50	0.39	0.90	0.56	1.45
Clusters	142	142	142	142	142
Observations	6827	6827	6827	6827	6827

Notes: This table presents the effects of each performance incentive treatment on teacher behavior as rated based on classroom videos. The unit of observation is at the classroom observation level. Teachers may be observed multiple times over the course of the intervention. Column (1) presents the average score on the CLASS rubric (Pianta et al., 2012), on a 7-pt scale, averaging across all 12 sub-dimensions. Columns (2)-(4) provide scores grouping the sub-dimensions into three main categories. Column (5) provides the number of minutes during the observation that were spent on testing or test-prep activities. All regressions include strata fixed effects and control for grade and video coder fixed effects. Values in parentheses are standard p-values. Values in brackets are randomization inference p-values, and values in curly brackets are q-values to adjust for false discovery rate. Standard errors are clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Effect of Teacher Time at Work

	Days present at school		Hours worked per day	
	(1) b/p/ri/qi	(2) b/p/ri/qi	(3) b/p/ri/qi	(4) b/p/ri/qi
Objective Treatment	2.426 (0.570) [0.618] {0.742}	1.554 (0.339) [0.392] {0.523}	0.262 (0.195) [0.318] {0.523}	0.293 (0.282) [0.319] {0.523}
Subjective Treatment	5.927* (0.0719) [0.0960] {0.523}	3.340*** (0.00947) [0.0100] {0.120}	0.0348 (0.840) [0.855] {0.855}	-0.0432 (0.832) [0.823] {0.855}
F-test pval (subj=obj)	0.300	0.149	0.134	0.121
Randomiz infer pval (subj=obj)	0.371	0.202	0.295	0.164
FDR q-val (subj=obj)	0.523	0.523	0.523	0.523
Control Group Mean	144.79	182.72	7.90	7.92
Clusters	295	277	295	277
Observations	6394	4363	6394	4363

Notes: This table presents the effects of each performance incentive treatment on teacher attendance and time at work. The outcome is the number of days present at work and the number of hours at work. Data comes from biometric clock in and out data collected at all schools. The restricted sample removes teachers who took long leaves of absence or only worked at the school system for one of the two terms. All regressions include strata fixed effects and control for baseline school average test score, grade and subject. Values in parentheses are standard p-values. Values in brackets are randomization inference p-values. Standard errors are clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Teacher’s beliefs about contract noise

	Perceived Contract Noise			
	Pooled	Raise out of my control	Effort increases raise (reverse)	Contract motivates me (reverse)
Subjective Treatment	-0.145*** (0.0550)	-0.117** (0.0530)	-0.131* (0.0697)	-0.188** (0.0734)
Observations	4535	4535	4535	4535
Objective Mean	0.15	0.12	0.13	0.19

Notes: This table summarizes teacher responses to questions about their contracts’ noisiness from the previous year at endline. The questions are asked in a scale of Strongly disagree (1) to Strongly agree (5). Responses are then converted in z-scores. Column 1 pools all three questions together. Column 2 presents results for the question, “Under this incentive contract, my raise is out of my control”. Column 3 is “Under this incentive contract, those who work harder earn more” (reverse-coded as the question implies lower noise), and column 4 is “This incentive contract is effective at motivating me” (reverse-coded as the question implies lower noise). Standard errors are clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: Teachers Perceptions about which Actions to Focus on by Treatment

	Admin (1)	Pedagogy (2)	Prof. Develop. (3)	Test Prep (4)
Subjective Treatment	0.0887* (0.0502)	-0.0175 (0.0498)	-0.0513 (0.0495)	-0.0623 (0.0497)
Observations	2887	2887	2887	2887

Notes: This table reports teachers' responses to a hypothetical scenario in which they are advising a teacher which actions they should take to increase their raise under a given treatment. Data was collected as part of the endline survey, and observations are at the unit of the teacher. Actions are categorized into four categories: administrative tasks, pedagogy, professional development, and test preparation. Table A5 provides teacher's weight for the full list of activities by treatment. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table 10: Instrumenting Noise with Manager Accuracy - First Stage

	Noise Index (z-score)			
	(1)	(2)	(3)	(4)
Manager rating inaccuracy (z-score)	0.133*** (0.0502)	0.123** (0.0502)	-0.316* (0.165)	-0.219** (0.106)
Subjective Treatment	-0.326*** (0.0626)	-0.116 (0.0867)	-0.887*** (0.180)	0.795 (0.528)
Subjective Treatment*Manager rating inaccuracy (z-score)	0.102* (0.0537)	0.103* (0.0537)	0.419** (0.178)	0.306** (0.120)
Sample	Teacher	Teacher	Student	Student
Distortion Controls		X		X
Control Group Mean	0.32	0.32	1.23	1.23
Clusters	290	290	245	245
Observations	3356	3356	436740	436740

Notes: This table presents the relationship between manager rating inaccuracy and teacher's rating of how noisy their contract was. The outcome is teacher's rating of how noisy their contract was as measured by an index of their response to the three questions shown in Figure 6. Columns (1) and (2) uses data at the teacher level. Columns (3) and (4) uses data at the teacher-student exam level. Student exam data is matched to all teachers who taught the student in the given exam subject for at least one term from January-December 2018. All regressions control for subject, class and manager inaccuracy squared. Columns (3) and (4) also control for school and student test baseline. Columns (2) and (4) add in additional controls to pick up other non-noise differences across contracts. These controls include weight placed on each of the four activity groups listed in Table 9, those values interacted with the Subjective treatment, when teachers said they learned about the treatment and how often they received information about the treatment. Standard errors are clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 11: Effect of Noise on Outcomes

	Hours worked		Endline Test Score		Socio-Emotional Score	
	(1)	(2)	(3)	(4)	(5)	(6)
Noise	-13.24 (10.10)	-12.93 (9.982)	-0.175** (0.0875)	-0.269** (0.121)	-0.0326 (0.132)	-0.205 (1.648)
Controls		X		X		X
AR test p value	0.05	0.06	0.05	0.02	0.785	0.746
Montiel-Pflueger effective first stage F stat	3.60	3.65	5.50	6.46	0.32	0.02
Control Group Mean	40.46	40.46	-0.02	-0.02	-0.00	-0.00
Clusters	290	290	245	245	317	317
Observations	3356	3356	436740	436740	56843	56843

Notes: This table presents the relationship between teacher's rating of the noisiness of their contract, instrumented by manager inaccuracy*Subjective Treatment, on teacher and student outcomes. Columns (1) and (2) use data at the teacher level. Columns (3) and (4) use data at the teacher-student exam level. Student exam data is matched to all teachers who taught the student in the given exam subject for at least one term from January-December 2018. Columns (5) and (6) uses data the student level. All regressions control for subject, class, subjective treatment, manager inaccuracy, and manager inaccuracy squared. Columns (3) and (4) also control for school and student test baseline. Columns (2), (4) and (6) add in additional controls to pick up other non-noise differences across contracts. These controls include weight placed on each of the four activity groups listed in Table 9, those values interacted with the Subjective treatment, when teachers said they learned about the treatment and how often they received information about the treatment. Standard errors are clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 12: Effect of Manager Preferences on Student Outcomes

	Test Scores (SD)		Socio-Emotional Score (SD)	
	(1)	(2)	(3)	(4)
% of pts: Admin x Subj Treat	-0.0678 (0.0870)	-0.0631 (0.0895)	0.123 (0.0798)	0.119 (0.0786)
% of pts: Professional Development x Subj Treat	0.215** (0.0974)	0.226** (0.0971)	0.125 (0.0871)	0.114 (0.0875)
% of pts: Pedagogy x Subj Treat	-0.0186 (0.0700)	-0.0161 (0.0727)	0.0147 (0.0626)	0.0212 (0.0632)
% of pts: Testing x Subj Treat	0.0903 (0.0924)	0.0906 (0.0919)	-0.114 (0.0861)	-0.120 (0.0877)
Controls	X		X	
Control Group Mean	0.02	0.02	-0.00	-0.00
Observations	307827	307827	268261	268261
Clusters	212	212	177	177

Notes: This table presents the relationship between evaluation criteria interacted with treatment on student outcomes. Data is at the student-test or student-teacher level. All regressions control for the four categories of evaluation criteria and subjective treatment and the basic controls included in tables 3 and 4. Columns (2) and (4) add in additional controls to pick up other non-distortion differences across contracts. These controls include the noise index, beliefs about whether the contract affects teacher competition or favors certain teachers, when teachers said they learned about the treatment, how often they received information about the treatment, and all of these variables interacted with subjective treatment. Standard errors are clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

9 References

References

- Amoateng, Acheampong Yaw, Sabiti Ishmael Kalule, and Tim B. Heaton**, “Gender and changing patterns of political participation in sub-Saharan Africa : evidence from five waves of the Afrobarometer surveys,” *Gender and Behaviour*, 2014, 12 (3), 5897–5910. Publisher: IFE Centre for Psychological Studies (ICPS).
- ASER**, *Annual Status of Education Report Pakistan* 2019.
- Baker, George**, “Distortion and Risk in Optimal Incentive Contracts,” *The Journal of Human Resources*, 2002, 37 (4), 728.
- Bandiera, Oriana, Michael Carlos Best, Adnan Qadir Khan, and Andrea Prat**, “The Allocation of Authority in Organizations: A Field Experiment with Bureaucrats,” 2020, p. 88.
- Barlevy, Gadi and Derek Neal**, “Pay for Percentile,” *American Economic Review*, August 2012, 102 (5), 1805–1831.
- Biasi, Barbara**, “The Labor Market for Teachers under Different Pay Schemes,” *American Economic Journal: Economic Policy*, 2021, 13 (3), 63–102.
- Bloom, Nicholas, Renata Lemos, Raffaella Sadun, and John Van Reenen**, “Does Management Matter in schools?,” *The Economic Journal*, 2015, 125 (584), 647–674. _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/econj.12267>.
- Brown, Christina**, “Understanding Gender Discrimination by Managers,” *Working paper*, 2025, p. 49.
- **and Tahir Andrabi**, “Inducing Sorting through Performance Pay: Experimental Evidence from Pakistani Schools,” *Working Paper*, July 2025.
- **, Supreet Kaur, Geeta Kingdon, and Heather Schofield**, “Cognitive Endurance as Human Capital,” *The Quarterly Journal of Economics*, 2025, p. qjae043.
- Bruhn, Miriam and David McKenzie**, “In Pursuit of Balance: Randomization in Practice in Development Field Experiments,” *American Economic Journal: Applied Economics*, October 2009, 1 (4), 34.
- Chetty, Raj, John N. Friedman, and Jonah E. Rockoff**, “Measuring the Impacts of Teachers II: Teacher Value-Added and Student Outcomes in Adulthood,” *American Economic Review*, September 2014, 104 (9), 2633–2679.

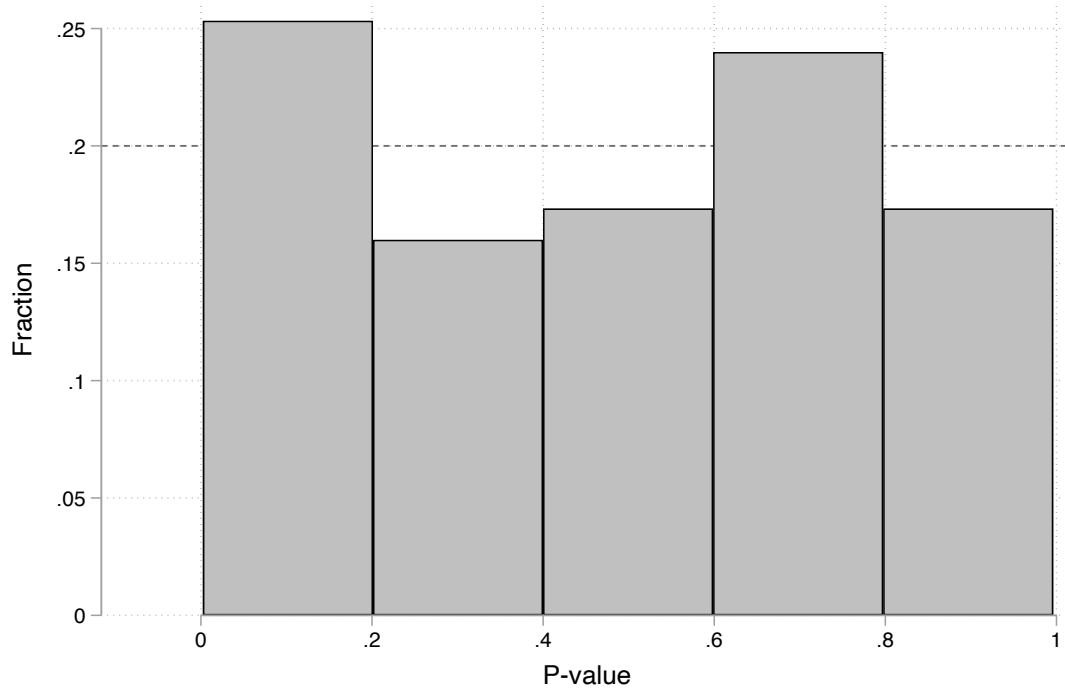
- de Janvry, Alain, Guojun He, Elisabeth Sadoulet, Shaoda Wang, and Qiong Zhang**, “Subjective Performance Evaluation, Influence Activities, and Bureaucratic Work Behavior: Evidence from China,” *American Economic Review*, 2023, 113 (3), 766–799.
- Deserranno, Erika, Philipp Kastrau, and Gianmarco León-Ciliotta**, “Promotions and Productivity: The Role of Meritocracy and Pay Progression in the Public Sector,” *American Economic Review: Insights*, 2025, 7 (1), 71–89.
- Engellandt, Axel and Regina Riphahn**, “Evidence on Incentive Effects of Subjective Performance Evaluations,” *Industrial & Labor Relations Review*, 2011, 64.
- Eren, Ozkan**, “Potential in-group bias at work: Evidence from performance evaluations,” *Journal of Economic Behavior & Organization*, 2023, 206, 296–312.
- Frederiksen, Anders, Fabian Lange, and Ben Kriechel**, “Subjective performance evaluations and employee careers,” *Journal of Economic Behavior & Organization*, 2017, 134, 408–429.
- Fryer, Roland G.**, “Teacher Incentives and Student Achievement: Evidence from New York City Public Schools,” *Journal of Labor Economics*, April 2013, 31 (2), 373–407.
- Gibbs, Michael, Kenneth A. Merchant, Wim A. Van der Stede, and Mark E. Vargus**, “Determinants and Effects of Subjectivity in Incentives,” *The Accounting Review*, 2004, 79 (2), 409–436.
- Goodman, Sarena F. and Lesley J. Turner**, “The Design of Teacher Incentive Pay and Educational Outcomes: Evidence from the New York City Bonus Program,” *Journal of Labor Economics*, April 2013, 31 (2), 409–420.
- Jackson, C. Kirabo**, “What Do Test Scores Miss? The Importance of Teacher Effects on Non-Test Score Outcomes,” *Journal of Political Economy*, October 2018, 126 (5), 2072–2107. Publisher: The University of Chicago Press.
- Jacob, Brian A. and Lars Lefgren**, “Can Principals Identify Effective Teachers? Evidence on Subjective Performance Evaluation in Education,” *Journal of Labor Economics*, January 2008, 26 (1), 101–136.
- Kashdan, Todd B., Melissa C. Stikma, David J. Disabato, Patrick E. McKnight, John Bekier, Joel Kaji, and Rachel Lazarus**, “The five-dimensional curiosity scale: Capturing the bandwidth of curiosity and identifying four unique subgroups of curious people,” *Journal of Research in Personality*, 2018, 73, 130–149.
- Khan, Adnan Q., Asim Ijaz Khwaja, and Benjamin A. Olken**, “Making Moves Matter: Experimental Evidence on Incentivizing Bureaucrats through Performance-Based Postings,” *American Economic Review*, January 2019, 109 (1), 237–270.

- Lavy, Victor**, “Using Performance-Based Pay to Improve the Quality of Teachers,” *The Future of Children*, 2007, 17 (1), 87–109.
- Lazear, Edward P and Paul Oyer**, “Personnel Economics,” in “The Handbook of Organizational Economics,” Princeton; Oxford: Princeton University Press., December 2012, pp. pp. 479–519.
- Leaver, Clare, Owen Ozier, Pieter Serneels, and Andrew Zeitlin**, “Recruitment, Effort, and Retention Effects of Performance Contracts for Civil Servants: Experimental Evidence from Rwandan Primary Schools,” *American Economic Review*, July 2021, 111 (7), 2213–2246.
- Leigh, A.**, “The Economics and Politics of Teacher Merit Pay,” *CESifo Economic Studies*, March 2013, 59 (1), 1–33.
- Muralidharan, Karthik and Venkatesh Sundararaman**, “Teacher Performance Pay: Experimental Evidence from India,” *Journal of Political Economy*, February 2011, 119 (1), 39–77.
- National Center for Education Statistics**, *Schools and Staffing Survey, 2010-2011: [United States]*, U.S. Dept. of Education, Office of Educational Research and Improvement, National Center for Education Statistics, 2011.
- Oyer, Paul and Scott Schaefer**, “Personnel Economics: Hiring and Incentives,” in O. Ashenfelter and D. Card, eds., *O. Ashenfelter and D. Card, eds.*, 1 ed., Vol. 4B, Elsevier, 2011, pp. 1769–1823.
- Pekrun, Reinhard, Thomas Goetz, Wolfram Titz, and Raymond P. Perry**, “Academic Emotions in Students’ Self-Regulated Learning and Achievement: A Program of Qualitative and Quantitative Research,” *Educational Psychologist*, 2002, 37 (2), 91–105. Publisher: Routledge.
- Pham, Lam D., Tuan D. Nguyen, and Matthew G. Springer**, “Teacher Merit Pay: A Meta-Analysis,” *American Educational Research Journal*, February 2020, 0 (0), 0002831220905580. _eprint: <https://doi.org/10.3102/0002831220905580>.
- Pianta, Robert C, Bridget K Hamre, and Susan Mintz**, *Classroom assessment scoring system: Secondary manual*, Teachstone, 2012.
- Podgursky, Michael J. and Matthew G. Springer**, “Teacher performance pay: A review,” *Journal of Policy Analysis and Management*, 2007, 26 (4), 909–950.
- Prendergast, Canice**, “The Provision of Incentives in Firms,” *Journal of Economic Literature*, 1999, 37 (1), 7–63.
- , “The Motivation and Bias of Bureaucrats,” *The American Economic Review*, 2007, 97 (1), 18.
- **and Robert Topel**, “Discretion and bias in performance evaluation,” *European Economic Review*, April 1993, 37 (2-3), 355–365.

- Rockoff, Jonah E. and Cecilia Speroni**, “Subjective and Objective Evaluations of Teacher Effectiveness,” *The American Economic Review*, 2010, *100* (2.), 261–266.
- Tang, Xin and Katariina Salmela-Aro**, “The prospective role of epistemic curiosity in national standardized test performance,” *Learning and Individual Differences*, 2021, *88*, 102008.
- The World Bank Group**, *Country Policy And Institutional Assessment Dataset* 2018.
- Wyk, Barend Van and Henry D Mason**, “The influence of learning and study strategies inventory on the success of engineering students at a South African University of Technology,” *Cogent Education*, 2021, *8* (1), 1933682. Publisher: Cogent OA _eprint: <https://doi.org/10.1080/2331186X.2021.1933682>.
- Yang, Huanxing**, “Efficiency Wages and Subjective Performance Pay,” *Economic Inquiry*, 2008, *46* (2), 179–196. _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1465-7295.2007.00069.x>.

A Appendix Figures and Tables

Figure A1: Distribution of P-Values from F-Test of Equality of Variance in Effort by Treatment



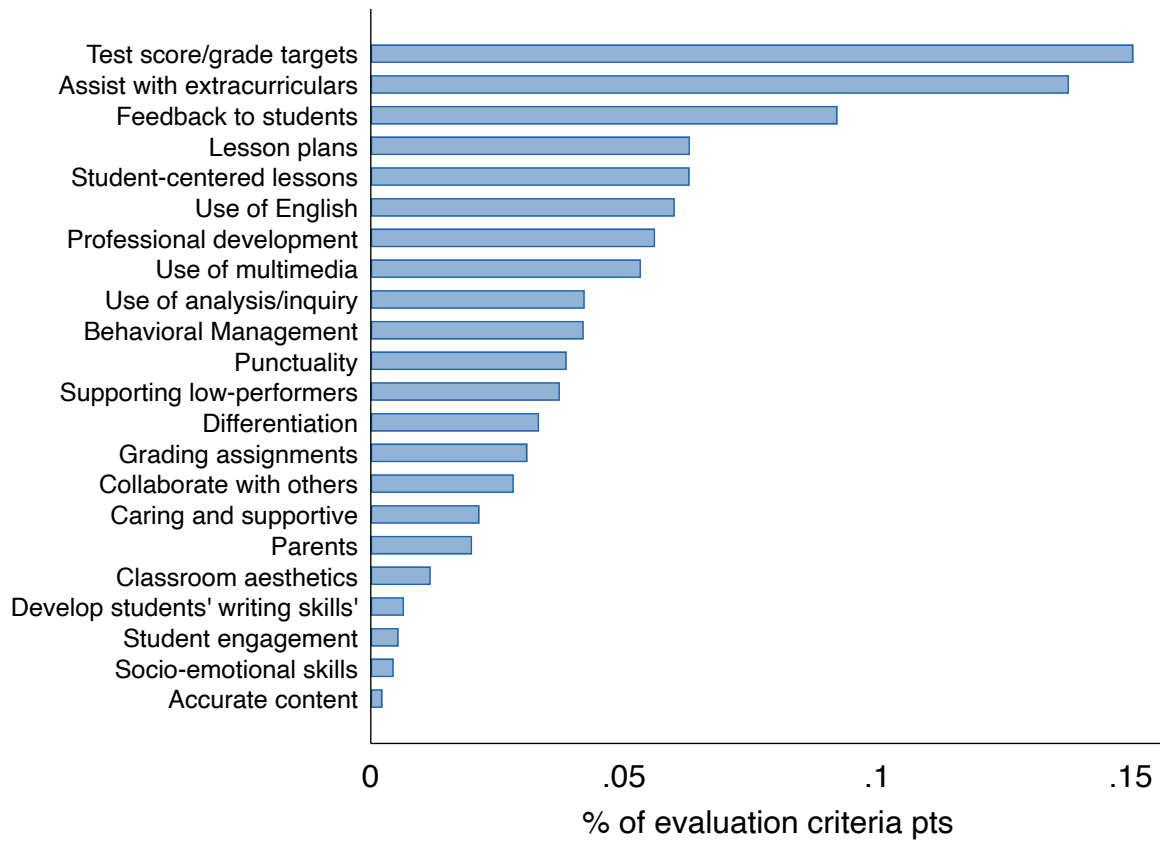
Notes: This figure plots the distribution of p-values for 75 F-tests. For each of our 25 different measures of teacher effort we calculate the within-school variance in effort along that metric and test for equality between each treatment pair (subjective = objective, subjective = control, objective = control). As the number of tests approaches infinity, for a true null of zero, we would expect p-values to be uniform at 0.2 (given five equal-width histogram bins).

Figure A2: Example Performance Criteria

PERFORMANCE APPRAISAL - FORM D			
Name:	Emp - 753 (43945)	Reporting to:	Emp - 19146 ()
Designation:	Teacher	School:	657 - North Nazimabad Primary III, Karachi
Employee Category :	Teaching Staff	Date of joining :	01/01/2013
Plan 1: Manager Appraisal of Effort			
Effort Criteria	Objective Score	Score Achieved	
Assessment of student understanding (monitoring of student learning, effective and timely copy checking)	20	20	
Differentiated lessons for varying learning needs	30	30	
Effectively delivering accurate and relevant content (effective implementation of the curriculum)	30	30	
Providing caring, supportive environment	20	20	
Total	100	100	

Notes: This figure shows an example set of performance criteria a teacher would have set in collaboration with their manager at the beginning of the year. This list of criteria was located on their employment portal, and available to access throughout the year. Managers could set individual criteria for each of their employees. These ranged from 4 to 10 criteria spanning numerous aspects of the teacher's job descriptions.

Figure A3: Percent of evaluation criteria points allocated to each category



Notes: This figure shows the percent of total points were allocated to evaluation criteria falling into one of 22 categories. Data on the text of the evaluation criteria and corresponding points comes from administrative employment records provided by the school system. The text responses were then categorized into categories by the research team.

Figure A4: Example Midterm Information

Dear Emp - 2890 ,

In keeping with the spirit of transparency and openness, we want to provide you with additional information about your performance this last term. We hope you'll use this information to continue to improve your practice. In addition, hopefully this information gives you an accurate picture of your progress up until this point and what you are currently on track to receive in your end of term appraisal.

As you know, similar to in past years your increment is based on your manager's appraisal of your performance. The change this year is that rather than the rating being based on your objectives and core competencies your rating will be based on your effort along several criteria.

These criteria are:

Effort Criteria	Total Points Possible
Assessment of student understanding (monitoring of student learning, effective and timely copy checking)	20
Differentiated lessons for varying learning needs	30
Effectively delivering accurate and relevant content (effective implementation of the curriculum)	30
Providing caring, supportive environment	20

Your midterm performance is:

Unsatisfactory	Satisfactory	Good	Very Good	Excellent

Ok

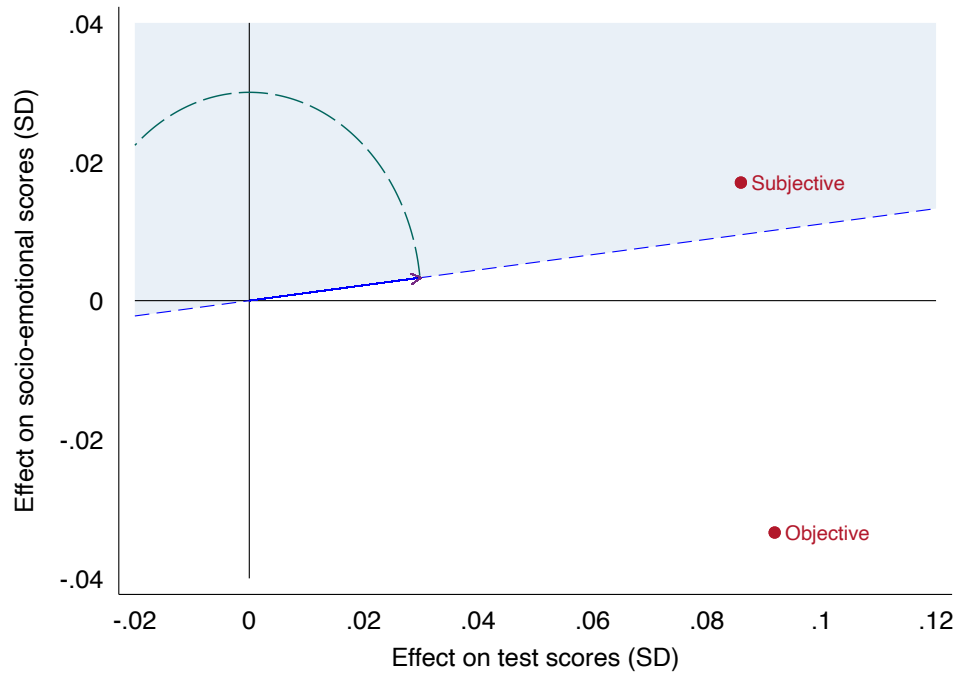
Notes: This figure shows an example notification sent to teachers during the summer between the two school years. The notification gave teachers a preliminary performance rating based on the first term of the experiment. Teachers received this information via email and as a pop-up notification on their employment portal. The example figure shows the notification that subjective treatment teachers would receive. Teachers in the objective treatment received midterm performance information based on their students percentile value added from the first term. Teachers in the control schools were randomly assigned to receive information about either their performance along the subjective criteria set by their manager or their students' percentile value added.

Figure A5: CLASS Observation, Example Rubric Dimension

Indicator	Behavioral Markers	Low (1, 2)	High (6, 7)
Relationships	- Physical proximity - Peer interactions - Shared positive affect - Social conversation	Teacher and students appear distant and disinterested	Teacher and students share warm and supportive relationship
Positive Affect	- Smiling - Laughter - Enthusiasm	Teacher and student do not appear to enjoy time in class	Frequent displays of positive affect
Positive Communication	- Positive comments - Positive expectations	Teacher and student rarely provide positive comments	Frequent positive communication among teachers and students
Respect	- Respectful language - Use of names - Warm, calm voice - Listening to others - Cooperation	Teacher and students rarely demonstrate respect for one another	Teacher and students consistently demonstrate respect for one another

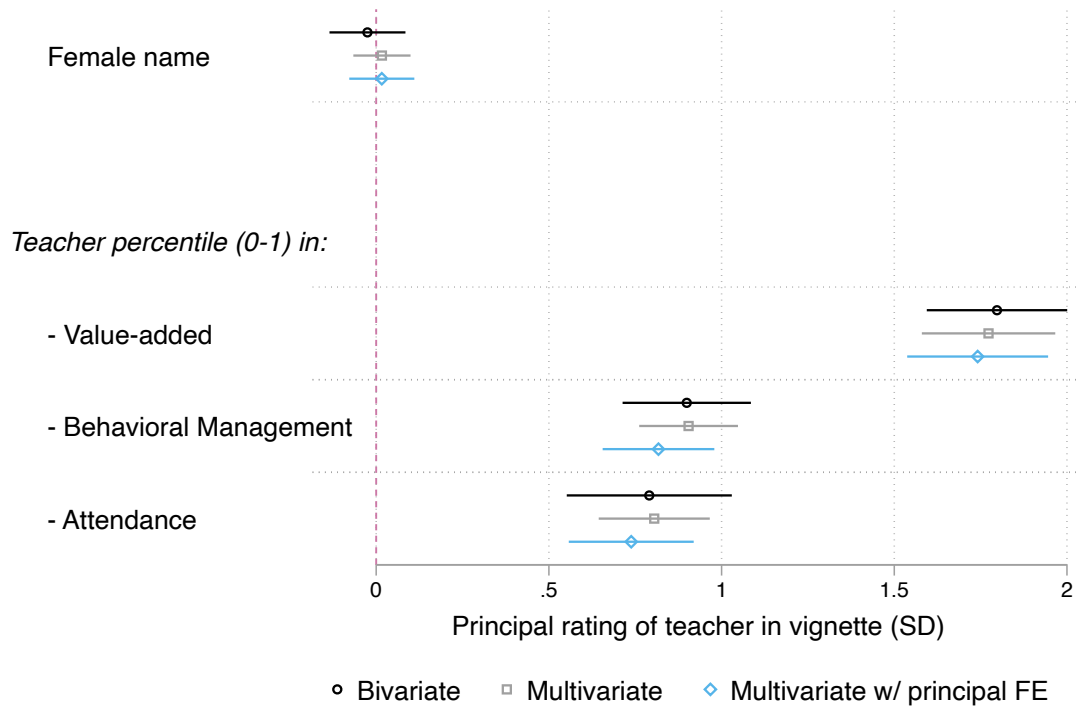
Notes: This figure shows the classroom observation rubric for one of the 12 dimensions of the CLASS rubric.

Figure A6: Total effect of performance pay on test scores and socio-emotional scores



Notes: This figure plots the treatment effect coefficients of the Subjective and Objective treatments (as shown in table 3 and table 4) where the x-axis is the effect on test scores in standard deviations and the y-axis is the effect on socio-emotional scores in standard deviations. The blue dotted line shows the set of preferences (weighting of effect on test scores versus socio-emotional scores) in which a person would be indifferent between the Subjective versus Objective treatments.

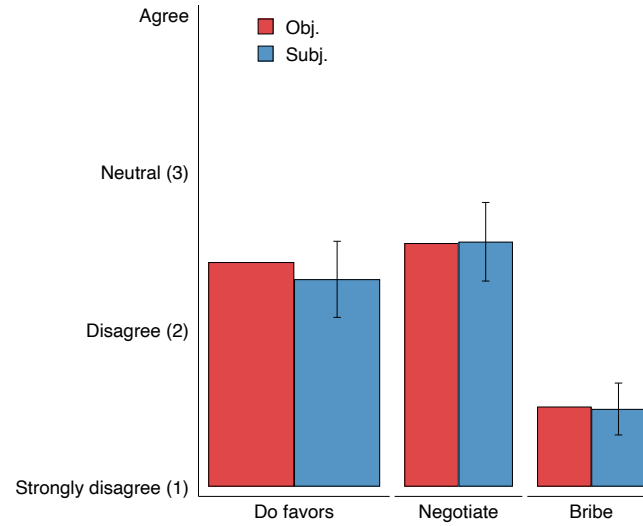
Figure A7: Manager Rating by Vignette Characteristics



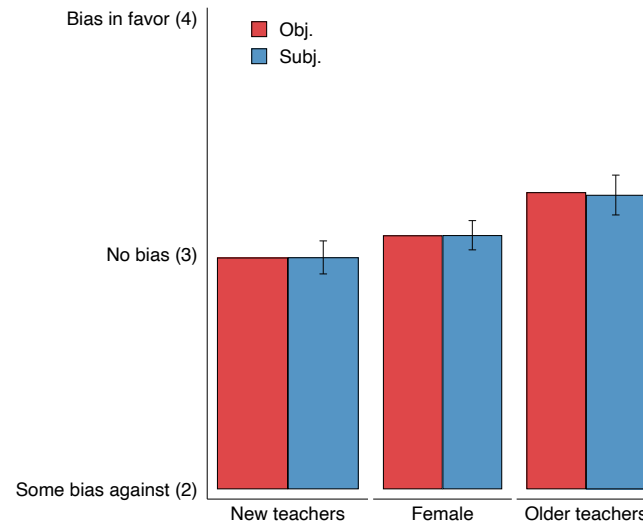
Notes: This figure presents the relationship between vignette characteristics and principal's rating of the teacher in the vignette. Principals received the vignettes stating, "[Female name/Male name] is in the [bottom/middle/top] 10% of teachers in terms of students' test score growth, in the [bottom/middle/top] 10% of teachers in terms of behavioral management, and is in the [bottom/middle/top]10% in terms of attendance and timeliness at work." Managers rated three such vignettes with characteristics randomized across vignettes. *Teacher Value-Added Percentile*, *Teacher Behavioral Management Percentile*, and *Teacher Attendance Percentile* takes values, 10, 50 and 90 to correspond to the bottom, middle and top 10% listed in the vignette. *Principal rating of teacher in vignette (SD)* is the principal's stated rating of the teacher in the vignette in standard deviations.

Figure A8: Beliefs about Contracts by Treatment

Teachers _____ to get a higher raise (scale 1-5)



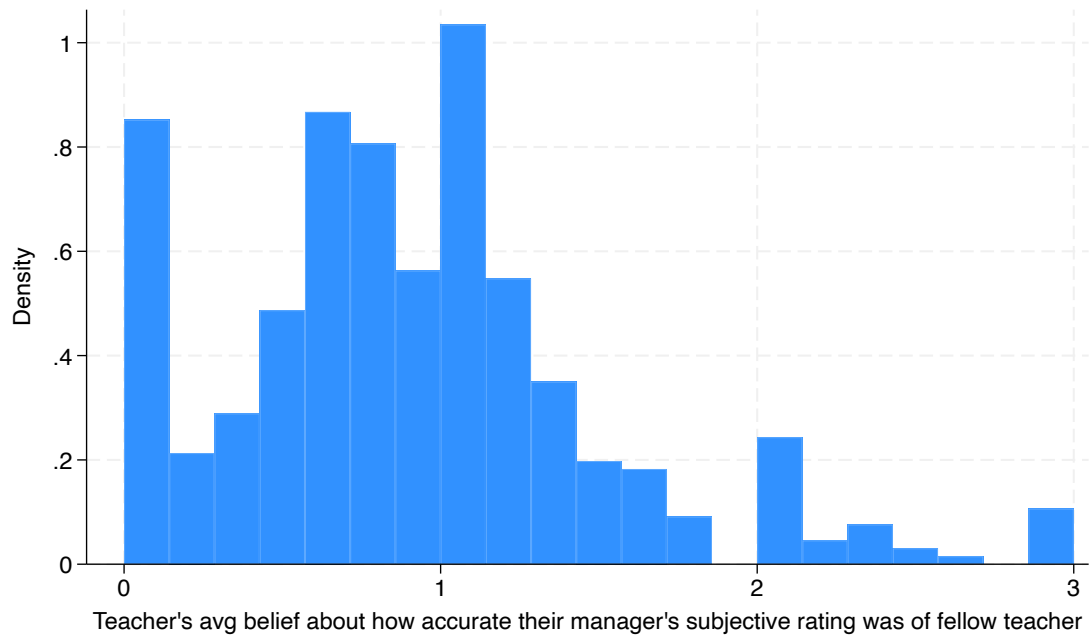
Is there any bias in favor or against: (scale: 1-5)



Notes: This figure presents teacher's responses to questions regarding their incentive contract for the previous year.

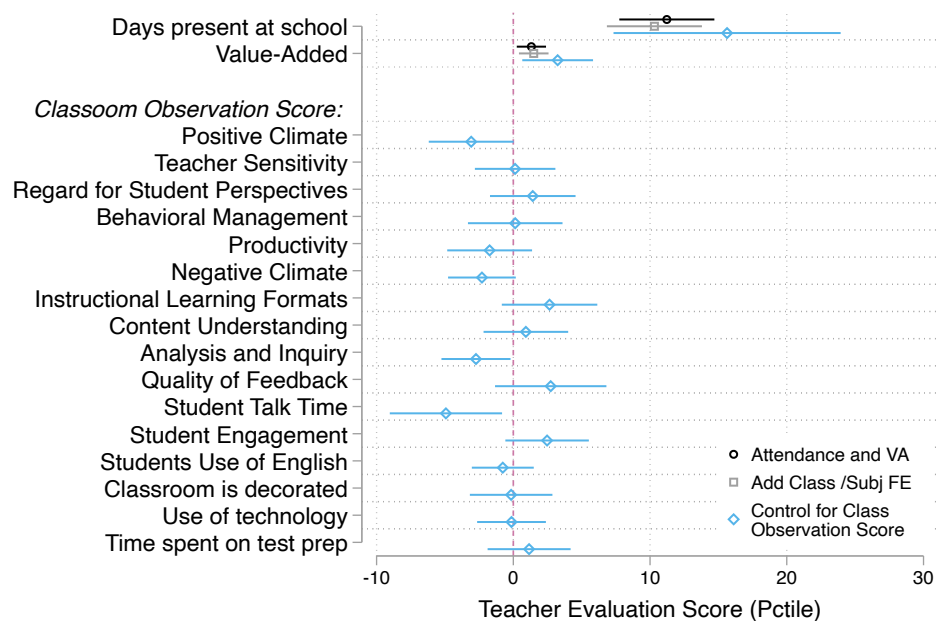
- Figure A shows teachers responses to questions about what “gaming” behaviors fellow teachers take to increase their raise. Figure B shows their responses to questions about whether certain groups are favored by the incentive scheme.
- The red (blue) bars presents the average response for teachers in the objective (subjective) treatment schools. The observation is at the teacher level and come from the endline survey of teachers.
- In figure A, the outcome is a 5pt scale from Strongly Disagree (1) to Strongly Agree (5). In figure B, the outcome is a 5pt scale (1, lots of bias against, 3, no bias, 5, lots of bias in favor).
- Standard errors are clustered at the school level. 95% confidence intervals are shown on the subjective bar comparing it to the objective treatment. Stars just above the bar show the significance of the subjective group relative to the objective. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A9: Distribution of Manager Perceived Inaccuracy Variable



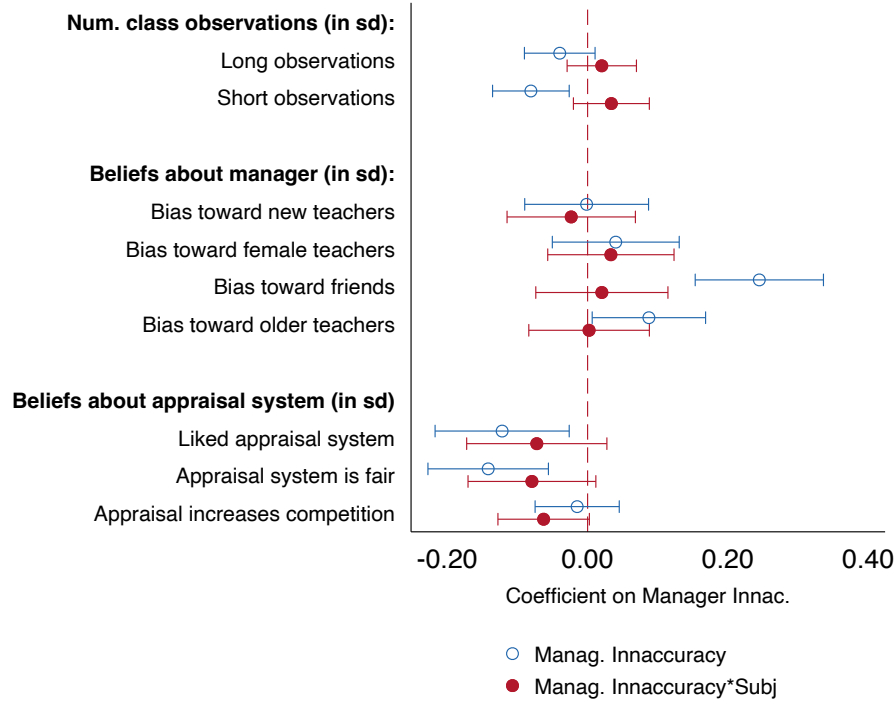
Notes: This figure plots the distribution of the manager inaccuracy variable. The variable uses data from the endline teacher survey asking teachers the following question about three fellow teachers in their school, “The appraisal score their manager would give them is... [Too high/low by more than one raise category], [Too high/low by about one raise category], [Too high/low by less than one raise category], or [Accurate]”. We then construct an average of these ratings per manager, capturing average *perceived* inaccuracy. A 0 means that for a given manager, across all their teachers surveyed, they believe their manager gave the exact correct subjective rating category. A score of 2 means, on average, a manager’s teachers believe that the manager rated other teachers too high/too low by 2 categories (out of the 5 possible category ranks).

Figure A10: Relationship between Teacher Behavior and Evaluation Score



Notes: This figure shows the relationship between teacher activity and performance evaluation score. Coefficients for a model only using teacher attendance and value-added to predict performance evaluation scores is shown in black circles. A model adding in class and subject fixed effects is shown in gray squares, and a model adding in our measures of classroom behavior from the dimensions of the CLASS rubric is shown with blue diamonds.

Figure A11: Management and Appraisal Practices by Noise Instrument



Notes: This figure presents teacher's endline survey responses about classroom observations, manager bias and beliefs about the performance incentive system by the instrument for noise (manager inaccuracy).

- The blue and red dots presents coefficient of a regression of the outcomes shown on the y-axis on *Manager Inaccuracy* and *Manager Inaccuracy*Subjective Treatment*. The regression also controls for all the same controls included in eq. 9. The observation is at the teacher level and come from the endline survey of teachers.
- All outcomes are converted into z-scores.
- Standard errors are clustered at the school level. 95% confidence interval bars shown * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A12: Management and Appraisal Practices by Distortion Measure

Fig. A: Points on Admin tasks

Fig. B: Points on Professional Development tasks

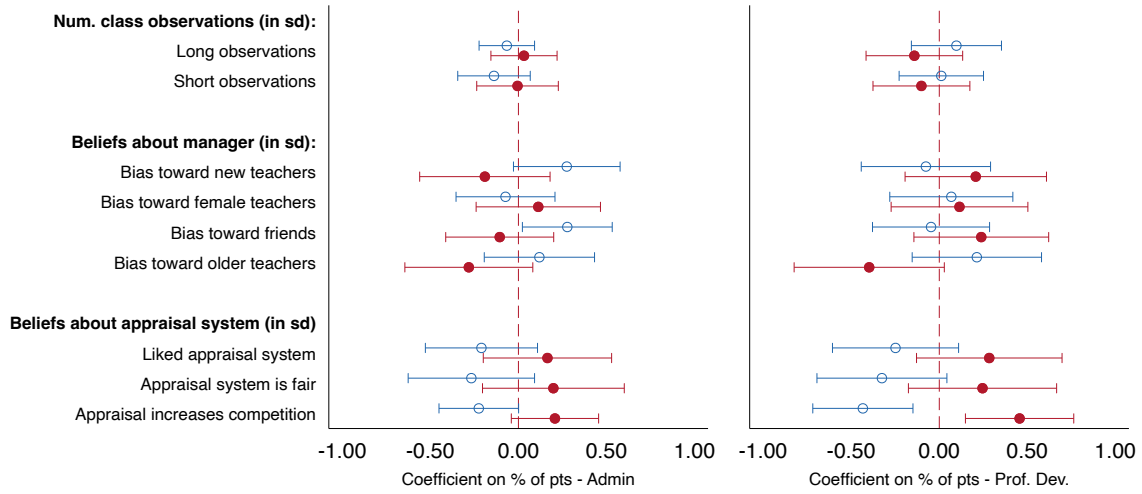
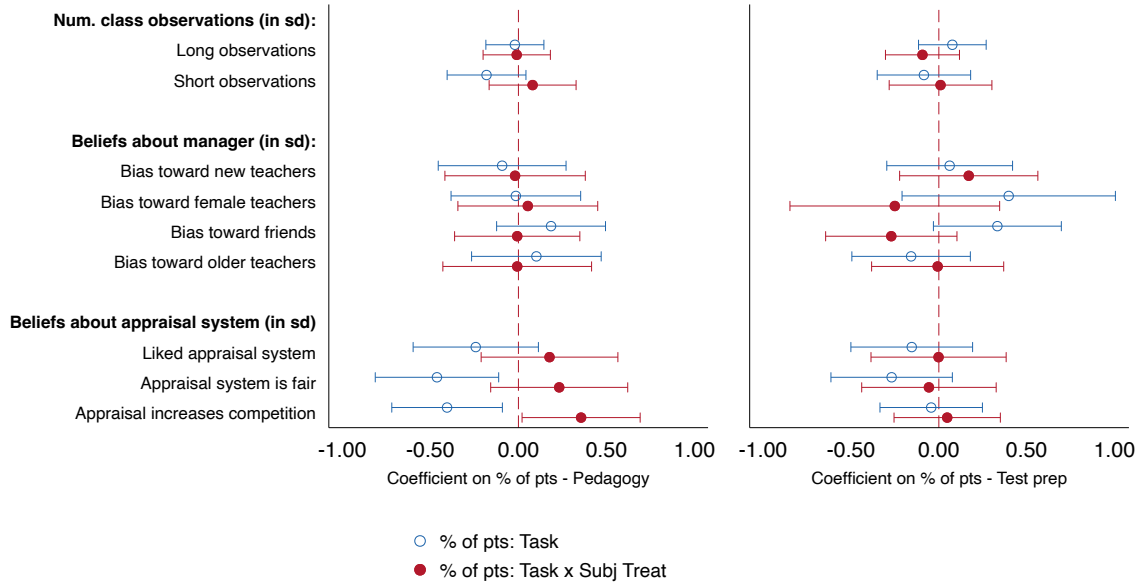


Fig. C: Points on Pedagogy tasks

Fig. D: Points on Test Prep tasks



Notes: This figure presents teacher's endline survey responses about classroom observations, manager bias and beliefs about the performance incentive system by our measure of distortion.

- The blue and red dots presents coefficient of a regression of the outcomes shown on the y-axis on *% of pts on a given task category* and *% of pts on a given task category*Subjective Treatment*, respectively. The regression also controls for all the same controls included in eq. 10. The observation is at the teacher level and come from the endline survey of teachers.
- All outcomes are converted into z-scores.
- Each figure corresponds to a different task category for teachers to be scored on: administrative tasks (fig. A), professional development tasks (fig. B), quality of pedagogy (fig. C) and test preparation tasks (fig. D).
- Standard errors are clustered at the school level. 95% confidence interval bars shown * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A1: Socio-Emotional Outcomes Student Survey

Question	Category	Source
1. I enjoy my math/science/English/Urdu class	Love of learning	Achievement Emotions Questionnaire
2. When work is difficult, I either give up or study only the easy part (reversed)	Love of learning	Learning and Study Strategies Inventory
3. I get very easily distracted when I am studying or in class (reversed)	Love of learning	Learning and Study Strategies Inventory
4. I can spend hours on a single problem because I just can't rest without knowing the answer	Love of learning	Epistemic Curiosity questionnaire
5. I feel sorry for other kids who don't have toys and clothes	Ethical	Eisenberg's Child-Report Sympathy Scale
6. Seeing a child who is crying makes me feel like crying	Ethical	Bryant's Index of Empathy Measurement
7. It is ok if a student lies to get out a test they are worried about failing (reversed)	Ethical	Adapted from McCabe/ICAI Academic Integrity Surveys
8. The pressure to do well is very high, so it is ok to cheat sometimes (reversed)	Ethical	Adapted from McCabe/ICAI Academic Integrity Surveys
9. I am interested in public affairs	Global	Afrobarometer/World Values Survey
10. This world is run by a few people in power, and there is not much that someone like me can do about it (reversed)	Global	Afrobarometer
11. People who are poor should work harder and not be given charity (reversed)	Global	Afrobarometer
12. It is important to protect the environment even if this means we cannot consume as much today	Global	Afrobarometer
13. People from other places can't really be trusted (reversed)	Global	Afrobarometer
14. I am comfortable asking my math/science/Urdu/English teacher for help or support	Inquisitive	Learning and Study Strategies Inventory
15. I enjoy learning about subjects that are unfamiliar to me.	Inquisitive	Epistemic Curiosity questionnaire
16. I would like to change to a different school	Dislike school	Learning and Study Strategies Inventory

Notes: This table presents the student survey question items used to assess student socio-emotional skills. Students rated these questions on a 5-pt scale from Strongly disagree to Strongly agree.

Table A2: Baseline Covariates

Variable	(1) Control		(2) Objective Treatment		(3) Subjective Treatment		T-test		
	N/ [Clusters]	Mean/ SE	N/ [Clusters]	Mean/ SE	N/ [Clusters]	Mean/ SE	(1)-(2)	Difference (1)-(3)	(2)-(3)
<i>Panel A: Teacher Characteristics</i>									
Performance evaluation score	656 [40]	3.360 (0.030)	384 [32]	3.362 (0.039)	3566 [139]	3.338 (0.010)	-0.002	0.022	0.024
Salary (USD)	920 [40]	5417.984 (313.504)	535 [32]	5125.462 (295.013)	4928 [145]	5329.416 (124.042)	292.523	88.569	-203.954
Age	921 [40]	36.591 (0.738)	539 [32]	36.083 (0.846)	4926 [145]	36.630 (0.298)	0.507	-0.039	-0.546
Years of experience	918 [40]	5.505 (0.277)	534 [32]	5.487 (0.425)	4897 [145]	5.725 (0.156)	0.019	-0.220	-0.238
<i>Panel B: Student Test Scores</i>									
Math Test Z-Score	9959 [40]	0.071 (0.070)	5292 [33]	-0.146 (0.065)	51775 [137]	-0.014 (0.026)	0.217**	0.085	-0.132*
Urdu Test Z-Score	9702 [40]	0.041 (0.072)	5259 [33]	-0.048 (0.063)	50915 [138]	-0.002 (0.028)	0.089	0.043	-0.046
English Test Z-Score	9755 [40]	0.017 (0.056)	5289 [33]	-0.049 (0.050)	51356 [137]	0.002 (0.032)	0.067	0.016	-0.051
Social Studies Test Z-Score	9171 [40]	0.041 (0.046)	5030 [33]	-0.064 (0.056)	49411 [137]	0.007 (0.022)	0.105	0.033	-0.071
Science Test Z-Score	9636 [40]	-0.010 (0.041)	5065 [33]	-0.064 (0.042)	50268 [137]	0.001 (0.024)	0.055	-0.011	-0.066

Notes: This table summarizes teacher and student characteristics before the experiment. The table reports mean values of each variable for each treatment group. The final three columns report mean differences between treatment group. Panel A presents teacher demographics as of September 2017. Panel B presents student test scores from yearly exams conducted in June 2017. Differences in sample size across teacher characteristics are due to some values being missing in the administrative data. For example, new teachers will not yet have a performance evaluation score at baseline. Differences in sample size in baseline student scores arise from either student absence on the day of the test or differences in student course selections in later secondary grades when students can choose to specialize. Standard errors are clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Effect of Incentives on Student Test Scores by Baseline Score

	Endline Test (z-score)	
	Student Baseline	School Baseline
Subjective Perf Pay	0.0944** (0.0386)	0.0956*** (0.0366)
Objective Perf Pay	0.0892* (0.0461)	0.0925* (0.0471)
Subjective Perf Pay* Covariate	-0.0262 (0.0220)	-0.0197 (0.0348)
Objective Perf Pay* Covariate	0.00139 (0.0184)	-0.00575 (0.0477)
F-test p-value (subj=obj)	0.90	0.94
F-test p-value (subj*covariate=obj*covariate)	0.23	0.73
Control Group Mean	-0.04	-0.04
Clusters	234	234
Observations	141566	141566
Controls		
Baseline	X	X
Grade/Subject	X	X

Notes: This table presents the heterogeneous effects of each performance incentive treatment on student endline test scores by student or school baseline scores (in SDs). The outcome is student's z-score on a given endline exam. The sample includes students tested in grades 4-13 in five subjects: Math, Science, English, Urdu, Economics. Column (1) uses the student's individual baseline test z-score as the interaction and column (2) uses the school mean baseline test z-score. All regressions include strata fixed effects and control for baseline student average test score, baseline school average test score, grade and subject. Values in parentheses are standard p-values. Standard errors are clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Effect of Incentives on Teacher Effort, Dropping First Observation

	CLASS Rubric				Test Prep
	All b/p/ri	Class Climate b/p/ri	Differentiation b/p/ri	Student-Centered b/p/ri	Minutes b/p/ri
Objective Treatment	-0.0741 (0.131) [0.183]	-0.0940* (0.0622) [0.100]	0.149** (0.0254) [0.0810]	-0.125** (0.0290) [0.0600]	0.575*** (0.00661) [0.0130]
Subjective Treatment	0.00969 (0.809) [0.814]	0.00157 (0.964) [0.973]	0.125** (0.0277) [0.0530]	-0.0171 (0.693) [0.738]	0.0871 (0.442) [0.734]
F-test pval (subj=obj)	0.059	0.047	0.695	0.037	0.013
Randomiz infer pval (subj=obj)	0.108	0.060	0.769	0.077	0.019
Control Group Mean	4.64	5.63	2.62	4.90	0.16
Clusters	139	139	139	139	139
Observations	5073	5073	5073	5073	5073

Notes: This table presents the effects of each performance incentive treatment on teacher behavior as rated based on classroom videos, similar to table 6, but drops the first hour of video when teachers are likely to exhibit the most sensitivity to being observed. The unit of observation is at the classroom observation level. Teachers may be observed multiple times over the course of the intervention. Column (1) presents the average score on the CLASS rubric (Pianta et al., 2012), on a 7-pt scale. Columns (2)-(4) provide scores on sub-areas of the class rubric. Column (5) provides the number of minutes during the observation that were spent on testing or test-prep activities. All regressions include strata fixed effects and control for grade and video coder fixed effects. Values in parentheses are standard p-values. Values in brackets are randomization inference p-values. Standard errors are clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Percent of Time Individuals Believe Should be Spent on Each Type of Activity

Variable	(1) Objective Teachers N/ [Clusters] Mean/ SE	(2) Subjective Teachers N/ [Clusters] Mean/ SE	(3) Subjective Managers N/ [Clusters] Mean/ SE	(1)-(2)	T-test Difference (1)-(3)	(2)-(3)
Improving behavioral management	487 (0.001)	2406 (0.001)	41 (0.006)	0.003**	0.009*	0.006
Collaborating with other teachers	487 (0.001)	2406 (0.000)	41 (0.005)	0.001	-0.008*	-0.009**
Grading student papers	487 (0.002)	2406 (0.001)	41 (0.005)	-0.003	-0.002	0.001
Providing differentiated lessons	487 (0.002)	2406 (0.001)	41 (0.005)	-0.003	0.000	0.003
Helping with extracurriculars	487 (0.002)	2406 (0.001)	41 (0.005)	-0.001	0.008	0.009
Incorporating higher order thinking skills	487 (0.002)	2406 (0.001)	41 (0.005)	0.001	-0.000	-0.001
Catering to different learning styles	487 (0.001)	2406 (0.001)	41 (0.005)	0.000	0.001	0.001
Incorporating multimedia	487 (0.001)	2406 (0.001)	41 (0.006)	-0.004**	-0.000	0.003
Communicating with parents	487 (0.001)	2406 (0.001)	41 (0.004)	0.002	0.001	-0.002
Conducting practice tests	487 (0.002)	2406 (0.001)	41 (0.007)	0.002	-0.001	-0.003
Making lessons more student centered	487 (0.001)	2406 (0.001)	41 (0.007)	-0.003**	-0.017***	-0.013***

Notes: This table reports teachers' responses to a hypothetical scenario in which they are advising a teacher which actions they should take to increase their raise under a given treatment. Data was collected as part of the endline survey, and observations are at the unit of the teacher/manager. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Manager Rating by Vignette Teacher Characteristic

	Manager Rating (z-score)					
	(1)	(2)	(3)	(4)	(5)	(6)
Value-Added Percentile	0.0180*** (0.00103)				0.0177*** (0.000979)	0.0174*** (0.00103)
Behavioral Management Percentile		0.00899*** (0.000941)			0.00904*** (0.000724)	0.00817*** (0.000819)
Attendance Percentile			0.00791*** (0.00121)		0.00805*** (0.000815)	0.00738*** (0.000917)
Female name				-0.0253 (0.0557)	0.0166 (0.0420)	0.0163 (0.0477)
Constant	-0.885*** (0.0738)	-0.451*** (0.0684)	-0.389*** (0.0731)	0.0128 (0.0634)	-1.731*** (0.106)	-1.639*** (0.0825)
Observations	567	567	567	567	567	567
Manager Fixed Effects						X

Notes: This table presents results from endline survey questions asking managers to rate a hypothetical teacher based on a description of their performance. The vignettes stated, “[Female name/Male name] is in the [bottom/middle/top] 10% of teachers in terms of students’ test score growth, in the [bottom/middle/top] 10% of teachers in terms of behavioral management, and is in the [bottom/middle/top]10% in terms of attendance and timeliness at work.” Managers rated three such vignettes with characteristics randomized across vignettes. *Teacher Value-Added Percentile*, *Teacher Behavioral Management Percentile*, and *Teacher Attendance Percentile* takes values, 10, 50 and 90 to correspond to the bottom, middle and top 10% listed in the vignette. *Teacher has female name* is a binary variable, which is 1 if the name used in the vignette is a traditionally female Pakistani name (Saadia, Haya, Maira, Anam, Zahra, or Sarah) and 0 if the name used is a traditionally male Pakistani name (Qasim, Tahir, Asim, Zain, Mujahid or Attefaq). Standard errors are clustered at the manager level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Effect of Incentives on Management of Teachers

	Panel A: Manager Observation Practices						
	Num long obs	Num short obs	Num walkthroughs	Improve	Distract	Freq	Appr
Subjective Treatment	-0.0180 (0.667)	-0.720 (1.105)	-0.0469 (1.526)	0.0918 (0.0583)	-0.0435 (0.0607)	0.00769 (0.0590)	0.117** (0.0580)
Objective Treatment	-0.359 (0.583)	-1.511* (0.910)	-0.440 (1.345)	0.151** (0.0725)	0.0657 (0.0697)	0.0630 (0.0718)	0.0548 (0.0766)
Observations	1588	1588	1588	1588	1588	1588	1588
F-test pval (subj=obj)	0.609	0.462	0.808	0.451	0.139	0.460	0.419
Control Mean	5.32	8.54	10.87	-0.11	0.05	-0.03	-0.14

	Panel B: Manager Hours Per Week Spent on:						
	Checking lesson plans	Class observ.	Mtg teachers/staff	Mtg parents	Extra-curric activ.	Monitoring duty	Paperwork
Subjective Treatment	-0.446 (1.520)	0.152 (1.860)	1.129 (0.785)	1.094 (1.162)	-0.298 (0.882)	0.897 (2.089)	1.328 (1.408)
Objective Treatment	-1.599 (1.444)	0.234 (2.846)	-1.853* (1.025)	-1.759 (1.264)	-0.872 (1.023)	0.0324 (2.064)	1.830 (2.887)
Observations	1255	1255	1255	1255	1255	1255	1255
F-test pval (subj=obj)	0.380	0.974	0.009	0.034	0.497	0.583	0.856
Control Mean	5.95	6.45	6.48	6.66	4.89	5.51	8.79

Notes: These tables presents the effects of each performance incentive treatment on manager's oversight of teachers. Panel A and B show effects on survey items from the teacher endline survey and manager endline survey, respectively. Columns 1-3 in panel A show the number of classroom observations the teacher said they received from their manager in the semester leading up to their annual performance evaluation (Aug-Dec). Columns 4-8 in panel A correspond to the following survey questions on a scale of Strong disagree (1) to Strongly agree (5) and then converted to a z-score.

- Improve: "Feedback from classroom observations allowed me to improve my teaching"
- Distract: "Observations add unnecessary stress and distract from my teaching"
- Freq: "Observations are a small snapshot of the classroom and are only effective if they happen frequently"
- Appr: "My appraisal was linked to the classroom observations my manager conducted"

All regressions include strata fixed effects. Values in parentheses are standard p-values. Standard errors are clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Teacher Effort and Subjective Performance Rating

	Subjective Performance Rating Percentile (0-100)		
	(1)	(2)	(3)
Hours present at school	-1.793*** (0.293)	-1.550*** (0.306)	-1.979*** (0.617)
Days present at school	0.167*** (0.0263)	0.153*** (0.0263)	0.232*** (0.0628)
Value-Added	1.393** (0.575)	1.574*** (0.581)	3.417** (1.388)
CLASS Rubric Dimensions:			
Positive Climate			-7.472* (3.836)
Teacher Sensitivity			0.323 (3.647)
Regard for Student Perspectives			1.650 (1.847)
Behavioral Management			0.282 (3.574)
Productivity			-3.829 (3.492)
Negative Climate			-12.49* (6.865)
Instructional Learning Formats			5.060 (3.396)
Content Understanding			1.780 (3.051)
Analysis and Inquiry			-4.815** (2.268)
Quality of Feedback			3.681 (2.791)
Student Talk Time			-5.721** (2.427)
Student Engagement			5.804 (3.651)
Other aspects of classroom observation:			
Students Use of English			-0.0498 (0.0747)
Classroom is decorated			-0.659 (6.348)
Use of technology			-0.0803 (0.780)
Time spent on test prep			0.978 (1.310)
Observations	2778	2628	618
Dependent Variable Mean	49.05	49.05	49.05
Subject and Grade Controls		X	X

Notes: This table presents the relationship between teacher behavior and their subjective performance rating. The dependent variable is subjective performance rating percentile. Column (1) and (2) includes the full sample of teachers and column (3) just includes teachers for whom we conducted a classroom observation. *Hours* and *Days present* are from biometric clock in and out data provided by the school system. Value-added is calculated using administrative test scores and endline test scores. The remaining variables are from classroom observations. The first 12 are the dimensions of the CLASS rubric and the rest are additional elements of teaching not captured by the CLASS rubric. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: Teacher's beliefs about contract features

Variable	(1) Objective Treatment N/[Clusters]	Mean/SE	(2) Subjective Treatment N/[Clusters]	Mean/SE	T-test Difference (1)-(2)
Panel A: Bias and Favoritism					
Is there any bias in favor or against the following groups (in the raises they receive)?					
New teachers	382 [33]	2.982 (0.029)	4237 [237]	2.983 (0.011)	-0.001
Female teachers	382 [33]	3.076 (0.029)	4237 [237]	3.077 (0.012)	-0.001
Older teachers	382 [33]	3.259 (0.054)	4237 [237]	3.248 (0.015)	0.011
Certain teachers are favored regardless of how hard they work	382 [33]	2.754 (0.050)	4237 [237]	2.772 (0.021)	-0.018
Panel B: Gaming					
Teachers do favors for managers to get a higher raise	124 [29]	2.427 (0.102)	2175 [208]	2.318 (0.038)	0.109
Teachers try to negotiate for a higher raise	124 [29]	2.548 (0.198)	2175 [208]	2.557 (0.037)	-0.009
Teachers bribe managers for a higher raise	124 [29]	1.508 (0.090)	2175 [208]	1.493 (0.026)	0.015
Panel C: Other features of the treatment					
How frequently did you think about the appraisal system	382 [33]	3.463 (0.149)	4237 [237]	3.479 (0.046)	-0.016
When did you come to understand what was expected under the contract	380 [33]	4.095 (0.128)	4199 [237]	4.089 (0.053)	0.006

Notes: This table summarizes teacher responses to questions about their contracts from the previous year at endline. The table reports mean values of each variable for objective versus subjective teachers. The final column reports mean differences between treatment group and report if any are statistically significant. The three "Is there any bias" questions are on a 5 pt scale (1, lots of bias against, 3, no bias, 5, lots of bias in favor). The remaining questions in panel A and B are on a 5-pt scale from 1 (strongly disagree) to 5 (strongly agree). Questions in panel C were on a scale from 1 to 8. Questions in panel B were asked to a 50% sub-sample of teachers in an effort to reduce overall survey length. Standard errors are clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A10: Heterogeneous Treatment Effects by Manager Characteristics

	Endline Test Scores					
	(1)	(2)	(3)	(4)	(5)	(6)
Subjective Treatment	-0.0156 (0.197)	0.169** (0.0688)	-0.0566 (0.117)	0.249*** (0.0775)	0.412 (0.681)	-0.0912 (0.491)
Interaction	0.00111 (0.00274)	0.00827 (0.00503)	0.0159 (0.0977)	0.142* (0.0763)	0.0386 (0.0863)	-0.0215 (0.0618)
Interaction*Subjective Treatment	0.00205 (0.00420)	-0.00883 (0.00648)	0.148 (0.127)	-0.211** (0.0910)	-0.0818 (0.162)	0.0375 (0.116)
Interaction	Age	Experience (years)	Female	Manager innacuracy (z-score)	Management Rating	Personnel Management Rating
Clusters	255	255	255	255	255	255
Observations	440595	440595	440595	440595	440595	440595

Notes: This table presents the treatment effects by manager characteristics. The row *Interaction* lists which characteristic is used as the interaction variable for a given column. Age, experience and gender are from administrative records. Manager inaccuracy is from teacher endline survey data. Mangement rating and Personnel management rating are from manager endline survey responses to World Management Survey questions. Standard errors are clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B Tournament versus Piece-Rate

Our conceptual framework assumes performance pay takes the linear piece rate form:

$$P(\mathbf{e}, \phi) = \mathbf{g} \cdot \mathbf{e} + \phi = g_1 e_1 + g_2 e_2 + \dots + \phi$$

The performance measure, $P(\mathbf{e}, \phi)$, is a function of teacher's effort, $\mathbf{e}$, and the marginal return to those actions on the performance measure, $\mathbf{g}$. In effect, $\mathbf{g}$ translates to an effective piece-rate for each effort category.

However, the actual performance scheme used in the experiment is based on within-school performance rank and with five rank categories:

Treatment: Performance Raise - Teachers receive a raise from 0-10% based on their within-school performance ranking using the percentile cutoffs below:

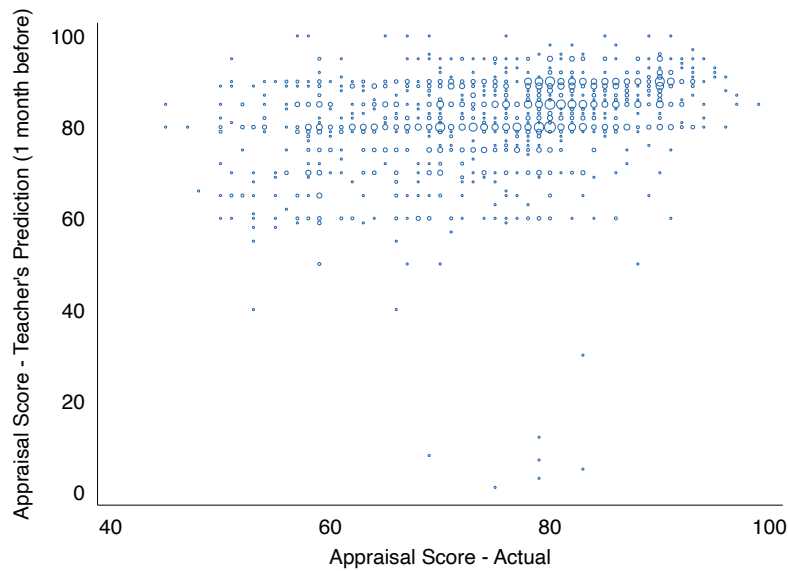
Performance Group	Within-School Percentile	Raise amount
Significantly above-average	91-100th	10%
Above-average	61-90th	7%
Average	16-60th	5%
Below average	3-15th	2%
Significantly below average	0-2nd	0%

There are several pieces of evidence from discussions with teachers and heterogeneous treatment effects results which suggest this first order linear approximation is appropriate.

A Difficulty in assessing position relative to performance cutoffs

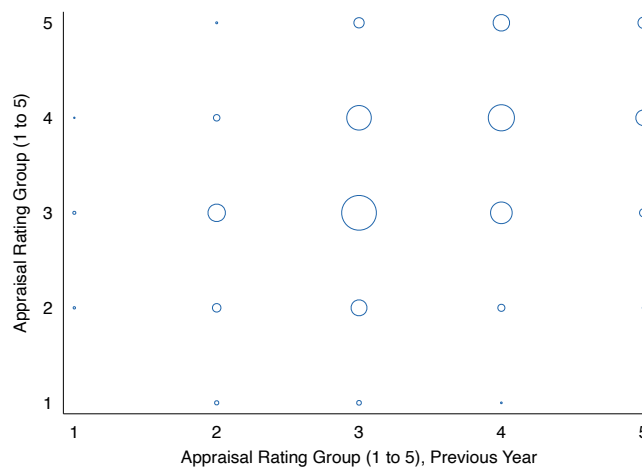
It is difficult for teachers to know exactly where they fall relative to the cutoff in the absence of additional effort and therefore what the return to effort will be. For teachers in the objective performance scheme they have very little information about their own ability and certainly do not have granular enough information to know exactly where they rank relative to their peers, so we will focus our evidence on whether teachers in the subjective treatment arms knew their position and acted strategically given the cutoffs

There are several pieces of evidence that support this idea. First, teachers are relatively incorrect in predicting what performance evaluation score they will receive in the following year. Most are overconfident in the score they will receive and only a very small fraction are within a point or two of their actual score received. The graph below shows actual subjective appraisal scores on the x-axis and teacher's predictions of what their score will be one month before it was announced.



Second, there is significant variation in the comparison pool year to year as overall teacher turnover is high, so the on average 29% of the teachers they will be compared with are new in the school. This makes it difficult for teachers to know exactly where they will fall.

Third, there is substantial year to year variation in appraisal score received, so there are large changes in score each year that would be hard for teachers to anticipate. The figure below shows the teacher's appraisal rating (from one to five) in a given year compared to the previous year. Only 46% of teachers received the same rating two years in a row and 7% of teachers receive a rating which is different by 2 or more steps from one year to the next. This makes it extremely difficult for teachers to be strategic about the threshold.

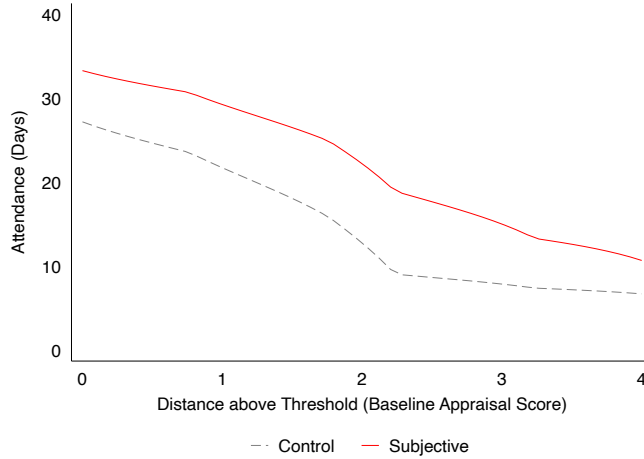


Finally, in conversations from teachers about how they responded to the incentives. In discussions

about how they changed their behavior in response to performance pay no one mentioned strategic behavior where they believed they were close to a certain performance cutoff.

B No differential effect by distance to cutoff at baseline

Finally, we can test this explicitly by showing whether there are heterogeneous effort responses by a teacher’s location relative to the threshold. We cannot know exactly the mental model a teacher has in mind for the counterfactual location they would be at without increasing effort. However, we can see the teachers subjective appraisal scores from the previous year and assume that is the reference point teachers have when deciding how much effort to exert. The graph below shows teacher attendance in the control and subjective treatment arm by number of points the teacher was above their current threshold. For teachers who are right at the bottom of their current threshold (0 or 1 pts), we might expect they would respond more strongly to the incentives for fear that they don’t get bumped down a category. However, we do not see a differential treatment effect by distance from the threshold in the previous year.



C Tournament Model Extension

Building off the model from section 2.1, we have:

The within-school tournament contract

Rather than a linear piece rate, the actual contract in the experiment is a within-school tournament with five rank-based pay categories. In a school with N core teachers, teacher i ’s bonus $B(r_i)$ is a weakly increasing step function of her within-school rank r_i on the performance measure

P_i :

$$B(r_i) = \sum_{k=1}^K b_k \cdot \mathbf{1}\{r_i/N \in [\tau_{k-1}, \tau_k)\}, \quad b_1 < b_2 < \dots < b_K, \quad (13)$$

with the cutoffs $\{\tau_k\}$ and prize levels $\{b_k\}$ identical across the subjective and objective treatment arms (see Appendix B for the exact schedule). The only difference between the two treatments is which performance measure P determines the ranking: the manager's subjective evaluation in one arm, and test-score value added in the other. Both arms have the same number of prize tiers, the same within-school relative nature, and the same cross-teacher variance in realized bonuses.

We study a symmetric pure-strategy equilibrium of this tournament. Let F_P denote the equilibrium distribution of P across teachers in a school and $h_k \equiv F'_P(P_{\tau_k})$ the density of P at the k -th cutoff. A teacher choosing effort $\mathbf{e}$ shifts her expected bonus through the effect of effort on the probability of clearing each cutoff. Standard tournament algebra (Lazear and Rosen, 1981; Green and Stokey, 1983) gives

$$\frac{\partial \mathbb{E}[B(r_i)]}{\partial e_{ik}} = g_k \cdot \underbrace{\sum_{k'=1}^{K-1} (b_{k'+1} - b_{k'}) h_{k'}}_{\equiv \beta(\sigma_P)}, \quad (14)$$

so that the tournament generates an effective piece rate $\beta(\sigma_P)$ on each effort dimension, scaled by the dimension-specific return g_k . The scalar β is the prize-weighted sum of order-statistic densities at the cutoffs and is a decreasing function of the dispersion of P across teachers, σ_P : as P becomes noisier, mass at any given cutoff falls and each additional unit of effort shifts a teacher's rank by less. This is where noise enters the first-order condition, and it does so because teachers face a relative-performance contract rather than a piece rate. Under a pure piece rate with risk-neutral agents, σ_ϕ^2 would not appear in the FOC at all; the tournament structure is what makes contract noise first-order relevant for effort.

Teacher's problem and equilibrium effort

Teachers have quadratic costs of effort around a baseline action vector $\vec{\eta}_i$ they would choose absent incentives:²⁶

$$u_i(\mathbf{e}) = \mathbb{E}[B(r_i)] - \sum_{k=1}^n \frac{(e_{ik} - \eta_{ik})^2}{2}. \quad (15)$$

²⁶Baker (2002) assumes risk-averse agents with utility $u(\mathbf{e}, \phi) = E[P] - r \text{var}[P] - \sum_i e_i^2/2$. Because we are not studying teacher retention, we omit the risk-aversion term, which only enters the participation constraint and does not change the effort FOC for a teacher who has accepted the contract.

Teachers are heterogeneous in $\vec{\eta}_i$ (their no-incentive baseline effort) but not in the cost slope. $\vec{\eta}_i$ is the action vector teacher i would choose in the control arm of the experiment, and can reflect cultural, professional, or institutional norms that vary across teachers. We do not assume $\vec{\eta}_i$ is fixed across treatments at the individual level; we only assume that its cross-sectional variance within a school is similar across treatments.

Taking the FOC of (3) using (14) and imposing symmetric equilibrium,

$$e_{ik}^* - \eta_{ik} = \beta(\sigma_P) g_k, \quad (16)$$

so equilibrium effort is the baseline plus the effective piece rate times the marginal return on the performance measure. Define $\Delta e_{ik}^* \equiv e_{ik}^* - \eta_{ik}$ as the effort response relative to the control arm. Expected student learning relative to the no-incentive baseline is then

$$\Delta V_i^* \equiv \mathbb{E}[V(\mathbf{e}_i^*, \epsilon)] - \mathbb{E}[V(\vec{\eta}_i, \epsilon)] = \beta(\sigma_P) \mathbf{f} \cdot \mathbf{g} = \beta(\sigma_P) \|\mathbf{f}\| \|\mathbf{g}\| \cos \theta, \quad (17)$$

where θ is the angle between $\mathbf{f}$ and $\mathbf{g}$.

Noise, distortion, and the two comparative statics

To express (17) in terms of the primitives we can measure in the data, take the variance of (2) in the symmetric equilibrium and assume the within-school covariance matrix of $\vec{e}$ is proportional to the identity:

$$\text{var}(P) = \|\mathbf{g}\|^2 \text{var}(\|\mathbf{e}\|) + \sigma_\phi^2.$$

Here $\text{var}(\|\mathbf{e}\|)$ is the cross-teacher variance in the norm of the effort vector within a school; it captures how much teachers within a school differ in total effort. It does not require that total effort be fixed in the absence of incentives. Our assumption that this quantity is similar across treatments follows from assuming homogeneous effort-cost slopes (teachers differ in $\vec{\eta}_i$ but respond to a given βg_k in the same way), and we test it directly in the data by comparing the within-school variance of 25 measured effort dimensions across treatment arms.²⁷

The tournament design also fixes $\text{var}(P)$ across the two treatment arms: the prize schedule is identical, and by construction the rank distribution (and therefore the cross-teacher variance of realized compensation) is the same in both arms. This is a feature shared by essentially any tournament or relative-performance system with a fixed prize structure. Under the maintained assumption that $\text{var}(\|\mathbf{e}\|)$ is also similar across treatments, differences in σ_ϕ^2 translate one-for-one into differences in $\|\mathbf{g}\|^2$: a noisier performance measure mechanically implies a smaller effective

²⁷We calculate the within-school variance in effort along 25 measured dimensions of teacher effort. Figure A1 plots the distribution of p-values for F -tests of equality across each treatment arm pair. The distribution is approximately uniform on $[0, 1]$, providing supporting evidence that $\text{var}(\|\mathbf{e}\|)$ is constant across treatment.

return to effort. Substituting into (17),

$$\Delta V_i^* = \beta(\sigma_P) \cdot \|\mathbf{f}\| \cdot \frac{\sqrt{\text{var}(P) - \sigma_\phi^2}}{\sqrt{\text{var}(\|\mathbf{e}\|)}} \cos \theta. \quad (18)$$

With $\|\mathbf{f}\|$, $\text{var}(P)$, and $\text{var}(\|\mathbf{e}\|)$ equal across the two treatments, the reduced-form difference in ΔV^* between the subjective and objective arms is driven by just two channels – the **noise** σ_ϕ^2 and the **distortion** $1 - \cos \theta$:

$$\frac{\partial \Delta V^*}{\partial \sigma_\phi^2} < 0 \quad (\text{student learning decreasing in contract noise}), \quad (19)$$

$$\frac{\partial \Delta V^*}{\partial (1 - \cos \theta)} < 0 \quad (\text{student learning decreasing in contract distortion}). \quad (20)$$

These are the two comparative statics we take to the data. The noise prediction is first-order because the contract is a within-school tournament: σ_ϕ^2 shrinks the effective piece rate $\beta(\sigma_P)$ and, equivalently, shrinks $\|\mathbf{g}\|$ given fixed $\text{var}(P)$. The distortion prediction is the standard Baker (2002) multi-tasking result and does not rely on the tournament structure. Finally, because subjective evaluation is a relative ranking by a manager whose beliefs about teacher effort need not coincide with the teacher's own, σ_ϕ^2 in the subjective arm has a natural structural interpretation as the variance of the gap between the manager's and the teacher's beliefs about $\mathbf{e}$. This maps directly to MacLeod's (2003), which shows that a decrease in the correlation between agent and principal beliefs lowers equilibrium effort – exactly the noise channel we study empirically using measured manager-accuracy heterogeneity.